

Chapter 6: Terpenoids

- **Unit description:**

- This unit deals with the classification, physicochemical properties, biosynthesis and significance of terpenoids in Pharmacy.

- **Unit learning outcomes:**

- Describe the basic chemistry of terpenes
- Use the isoprene rule in identifying isoprene units of terpenoids
- Classify terpenes as mono-, di- tri-, etc... by determining the number of isoprene units or as cyclic or acyclic
- Discuss general properties of terpenoids
- Describe the biosynthetic routes to terpenoids
- Explain pharmaceutical significance of terpenes

- **Contents:**

- 6.1. Introduction

- 6.1.1. Definition

- 6.1.2. Terpenoid nomenclature

- 6.2. The isoprene rule

- 6.3. Classification of terpenoids

- 6.3.1. Monoterpenes

- 6.3.2. Sesquiterpenes

- 6.3.3. Diterpenes

- 6.3.4. Triterpenes

- 6.3.5. Tetraterpenes

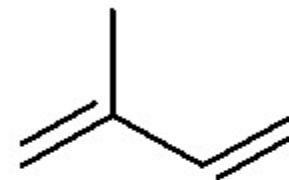
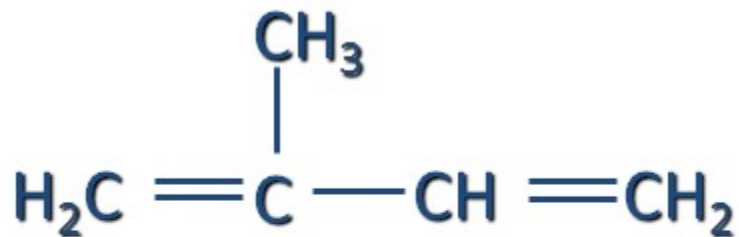
- 6.4. General properties of terpenoids

- 6.5. Biosynthesis of terpenes: An overview

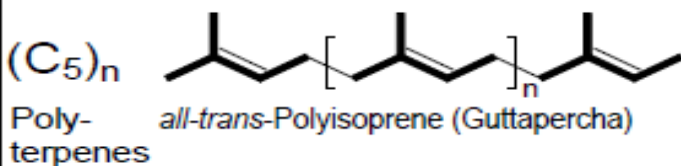
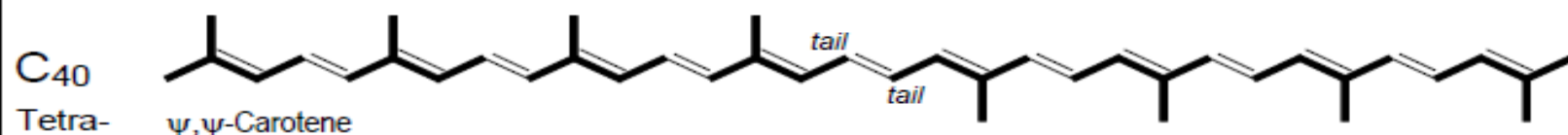
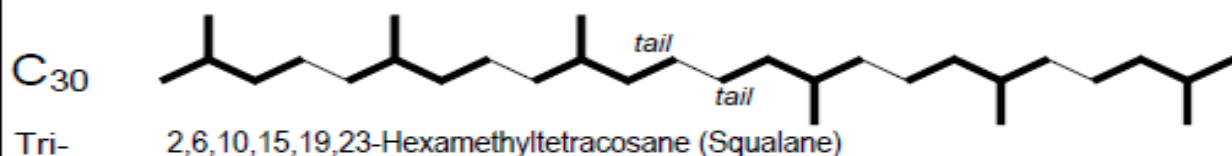
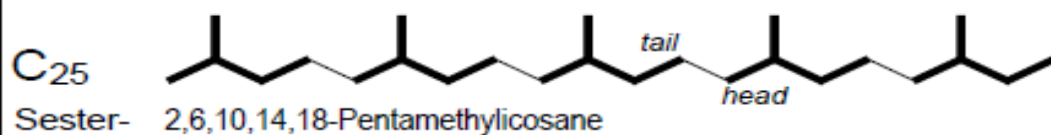
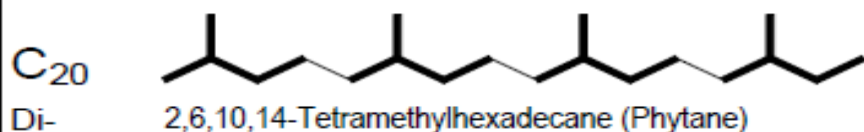
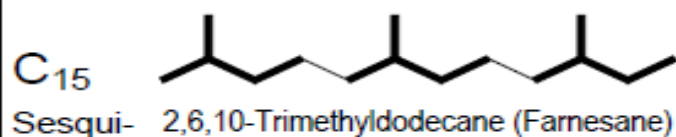
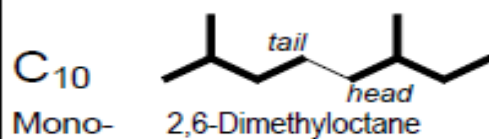
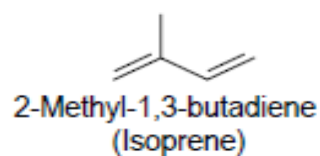
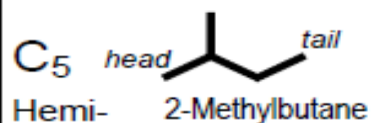
- 6.6. Significance of terpenoids in pharmacy

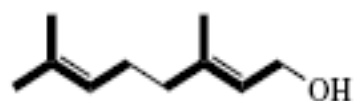
1. Definition

- Terpens/Terpenoids are large and structurally diverse family of natural products that are structurally related to isoprene
 - ✓ **Polymers of isoprene (2-methylbuta-1,3-diene)**

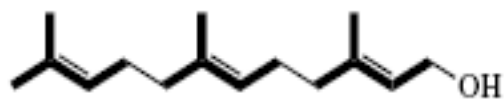


Isoprene
(2-methyl-1,3-butadiene)

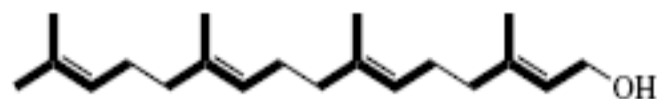




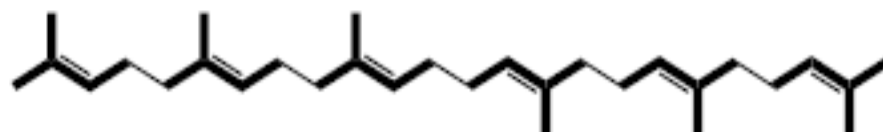
geraniol



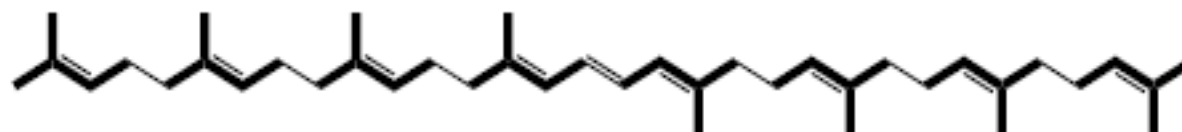
farnesol



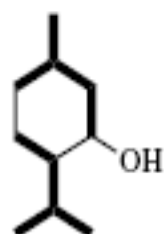
geranylgeraniol



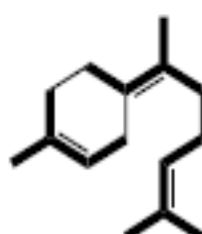
squalene



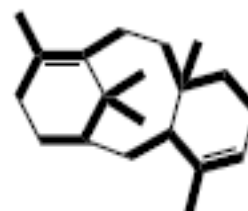
phytoene



menthol

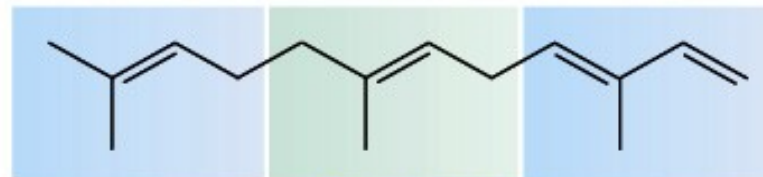
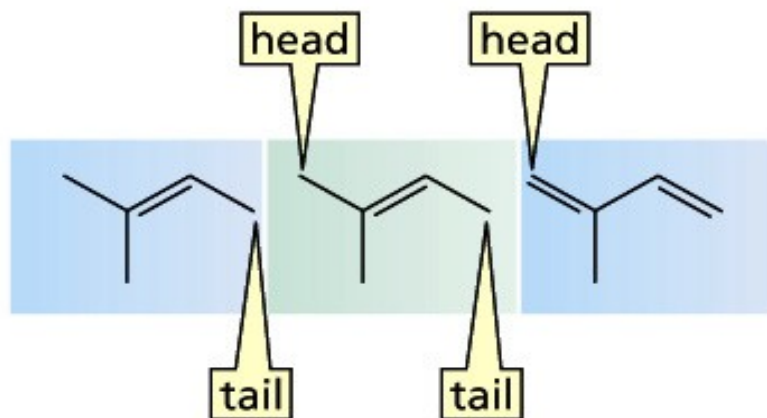


bisabolene



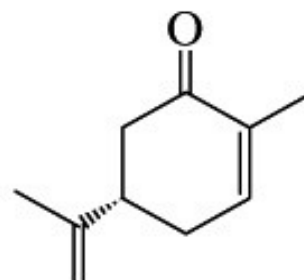
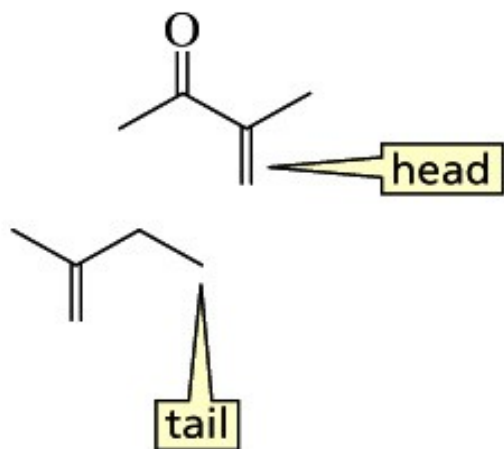
taxadiene

Examples:



α -farnesene

a sesquiterpene found in the waxy coating on apple skins



carvone

spearmint oil
a monoterpene

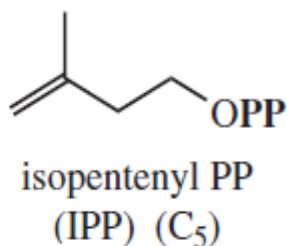
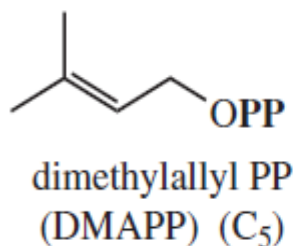
2. Classification

- ✓ **Based on the number of isoprene units**
- ✓ **Cyclic/ acyclic structures**
- ✓ **Presence/ absence of oxygen**
 - **terpenes: purely hydrocarbons, no oxygen**
 - **terpenoids: those containing oxygen in different forms**

Classification on the basis of the number of isoprene units

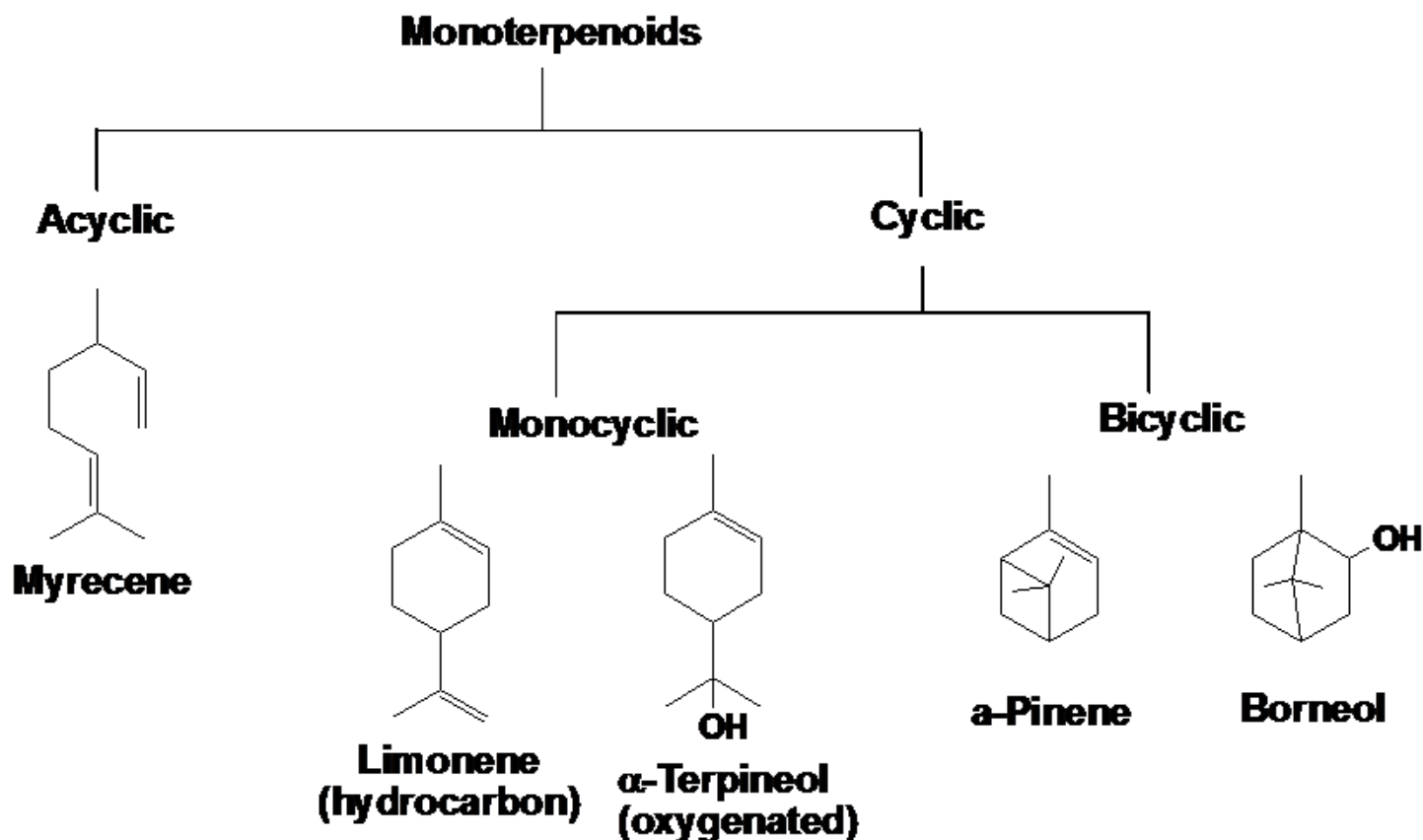
Hemiterpenes

- **IPP** and **DMAPP** are reactive hemiterpene intermediates in the pathways leading to more complex terpenoid structures
- They are also used as alkylating agents in the formation of meroterpenoids
- Relatively few true hemiterpenes are produced in nature
- **isoprene**, a volatile compound which is released by many species of plants, especially trees, being the notable example.
 - is formed by loss of a proton from the allylic cation



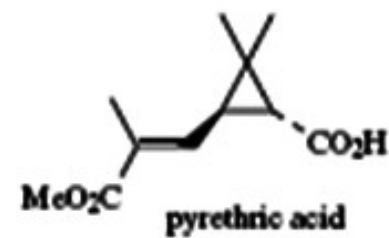
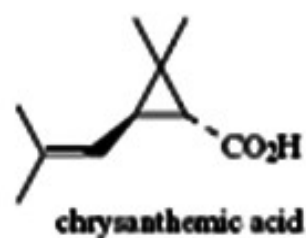
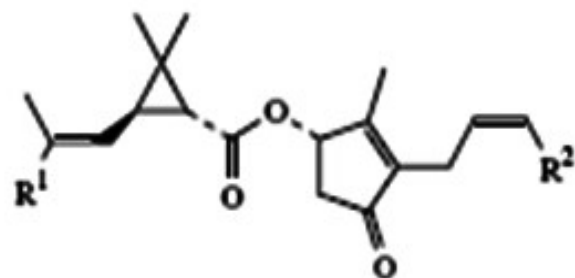
Monoterpenes

- contain two isoprene units linked head to tail
 - most are found free, some are glycosides
 - are volatile and constitute the major part of volatile oils

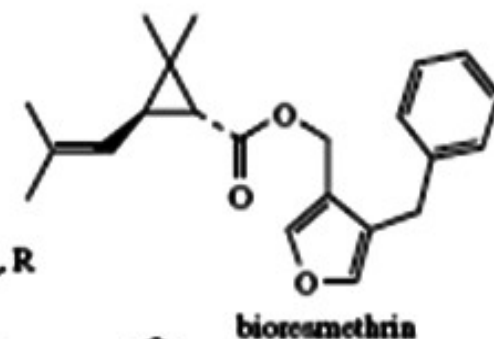
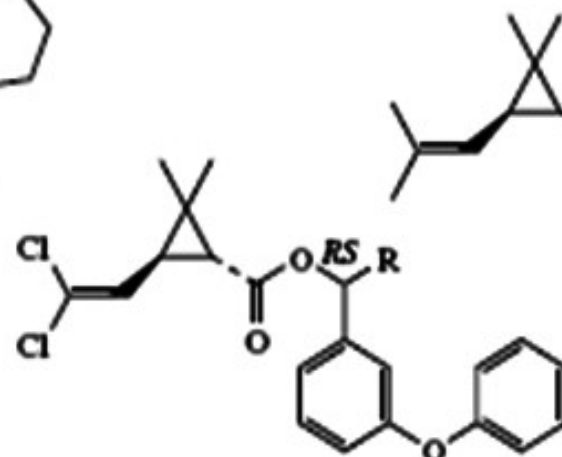
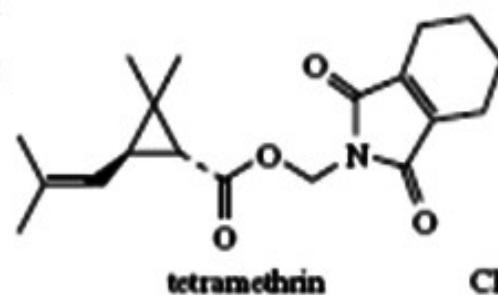
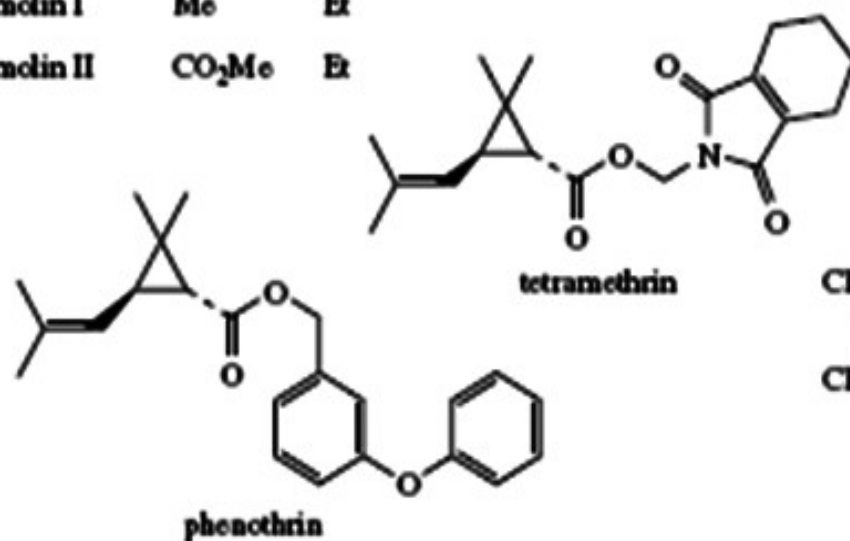
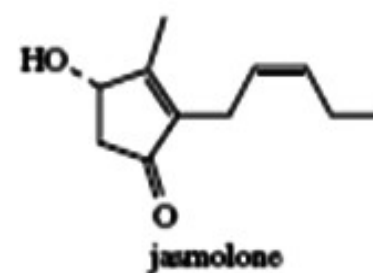
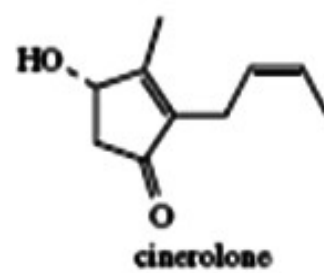
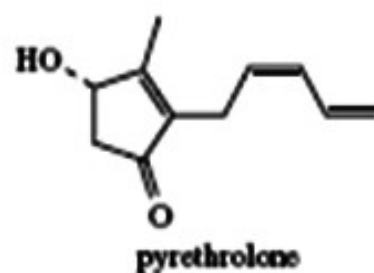


Irregular Monoterpenes

- Contain carbon skeletons derived from isoprene units,
 - do not seem to fit the regular head-to-tail coupling mechanism
 - termed irregular monoterpenes
- Seem to be limited almost exclusively to members of the plant family the Compositae/Asteraceae
- Although DMAPP and IPP are utilized in their biosynthesis, geranyl PP and neryl PP do not appear to be involved

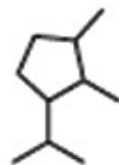


	R^1	R^2
pyrethrin I	Me	$\text{CH}=\text{CH}_2$
pyrethrin II	CO_2Me	$\text{CH}=\text{CH}_2$
cinerin I	Me	Me
cinerin II	CO_2Me	Me
jasmolin I	Me	Et
jasmolin II	CO_2Me	Et

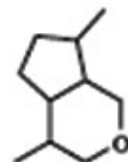


Iridoids

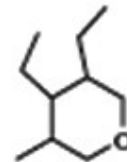
- The **iridane** skeleton found in **iridoids** is monoterpenoid in origin and contains
- a cyclopentane ring which is usually fused to a six-membered oxygen heterocycle
- A large number of iridoids are found as glycosides



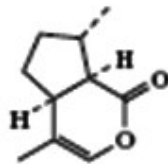
iridane



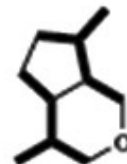
iridoid



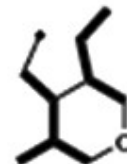
secoiridoid



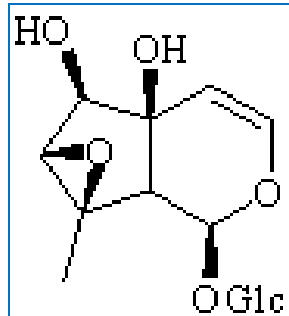
nepetalactone



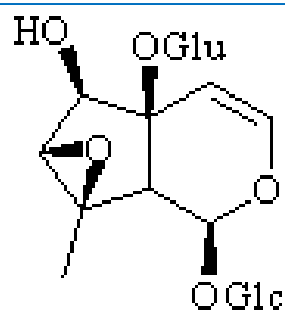
iridoid



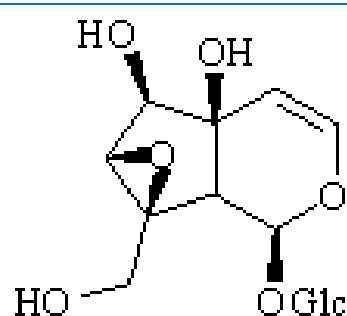
secoiridoid



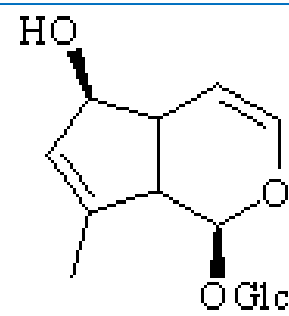
Antirrhinoside



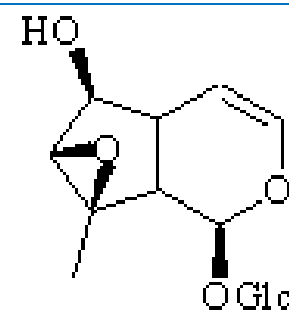
5-Glucosyl-
antirrhinoside



Mcfadienoside



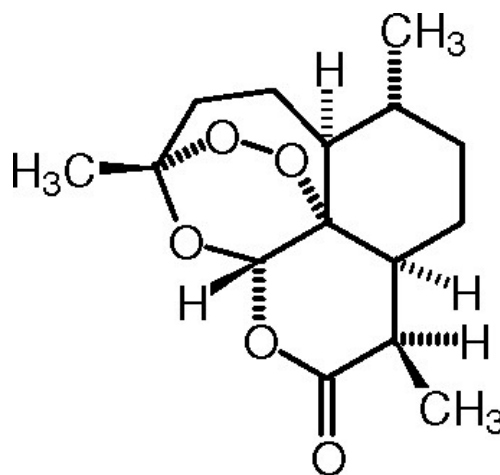
Linaride



10-Deoxycatalpol

Sesquiterpenes

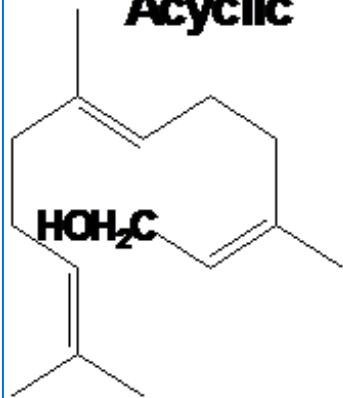
- contain 3 isoprene units linked head to tail
 - common components of volatile oils
 - Some contain lactone ring and are called sesquiterpene lactones (non-volatile)



artemesinin

Sesquiterpenes

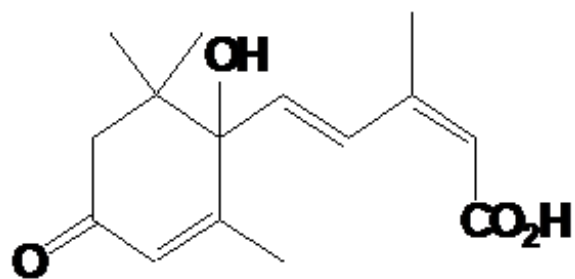
Acyclic



Farnesol

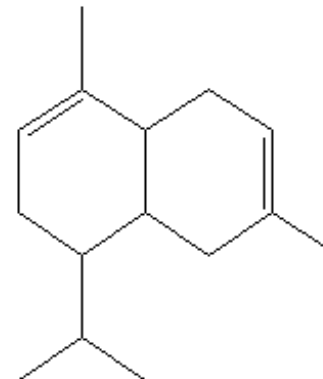
Cyclic

Monocyclic



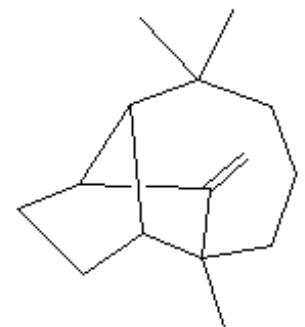
Absciscic acid

Bicyclic



Cadinene

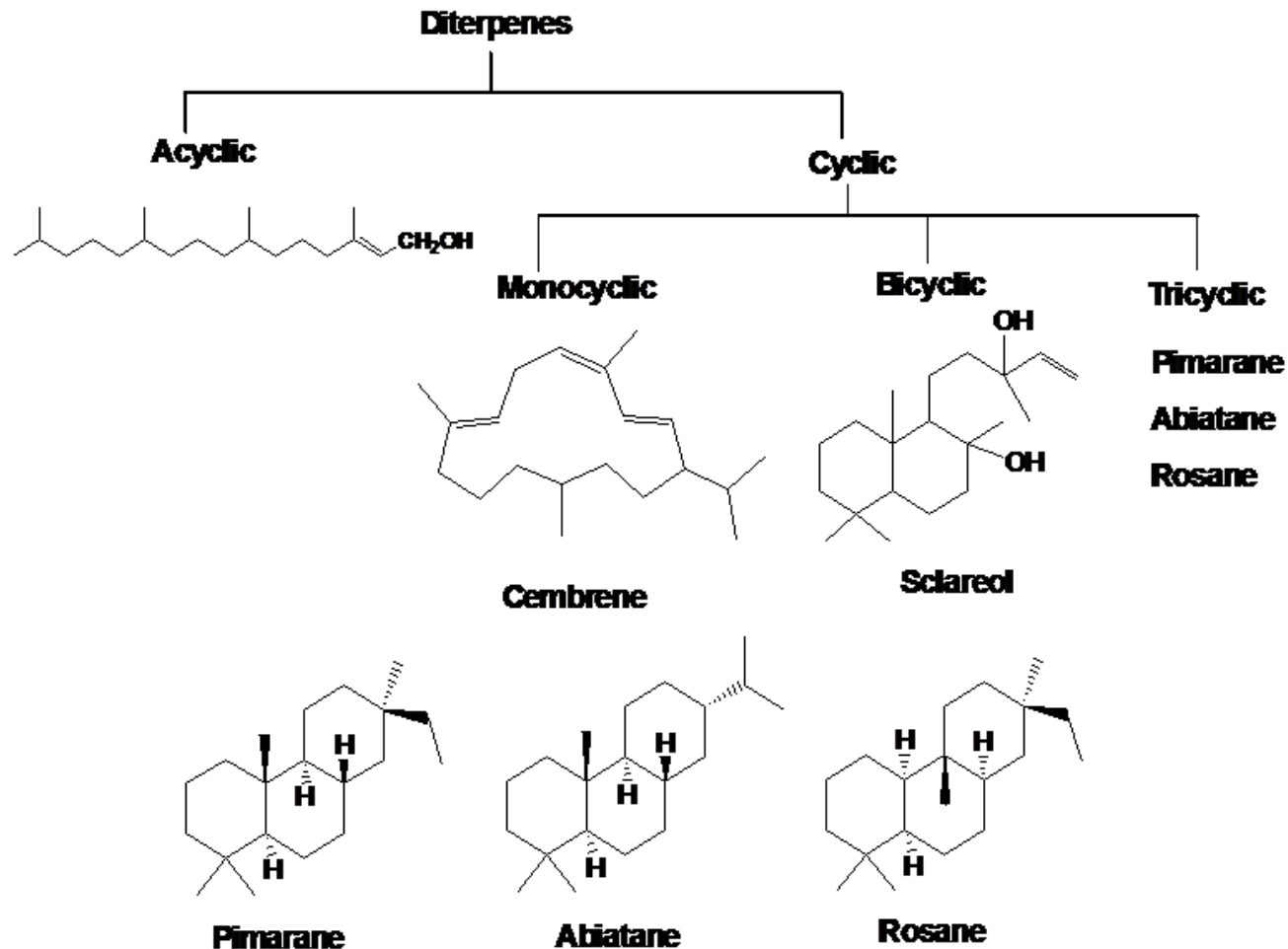
Tricyclic



Longifolene

Diterpenes

- contain 4 isoprene units linked head to tail
 - usual components of plant resins



Sesterterpenes

5 isoprene units they are mostly diterpenes with an extra isoprene unit

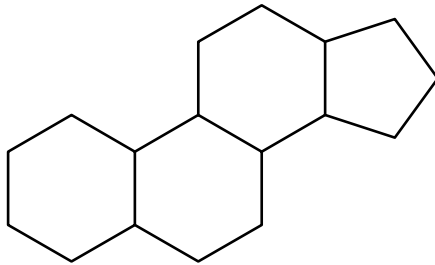
Triterpenes

- **contain 6 isoprene units**
 - **formed by tail to tail linkage of two sesquiterpene units**
 - **Rearrangement and/ or elimination of segments is very common giving rise to modified triterpenes**
 - **further classified into pentacyclic triterpenes and steroids**

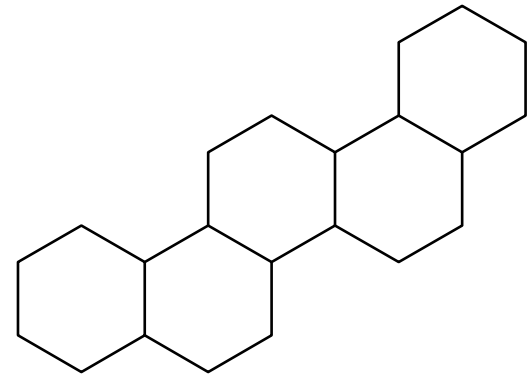
Triterpenes

Steroids

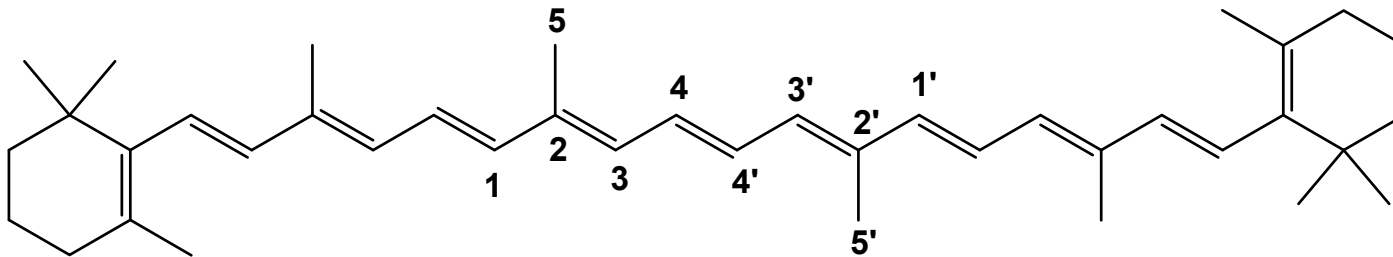
They are based on the
cyclopentanoperhydrophenanthrene ring



Pentacyclic triterpenoids



Tetraterpenes



β -Carotene

Tetraterpenes

- 8 isoprene units
 - formed by tail to tail linkage of two diterpene units

Polyterpenes

- poly-*cis*-terpenes rubber (1500-15000 isoprene units)
- poly-*trans*-terpenes gutta percha (~100 isoprene units)

Summary

Class	Number of carbon atoms
Monoterpene	10
Sesquiterpene	15
Diterpene	20
Sesterpene	25
Triterpene	30
Tetraterpene	40

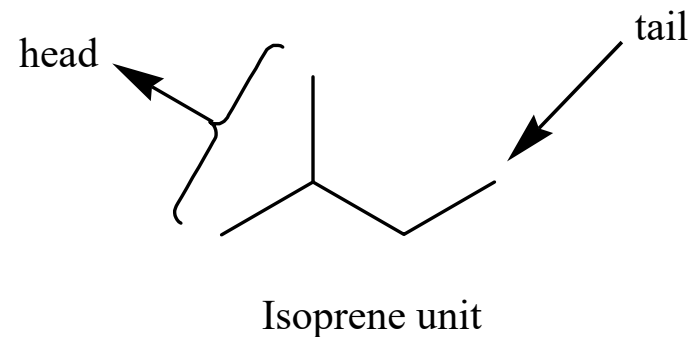
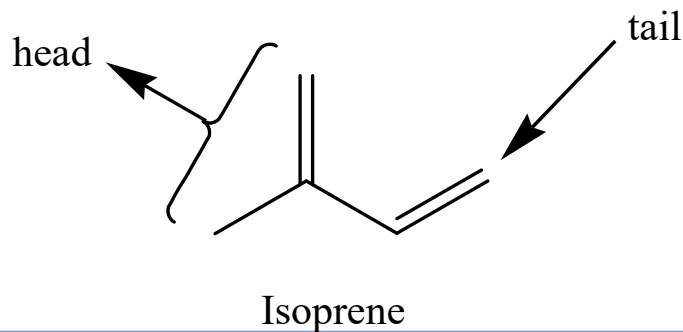
The isoprene rule

The isoprene rule

- The isoprene rule

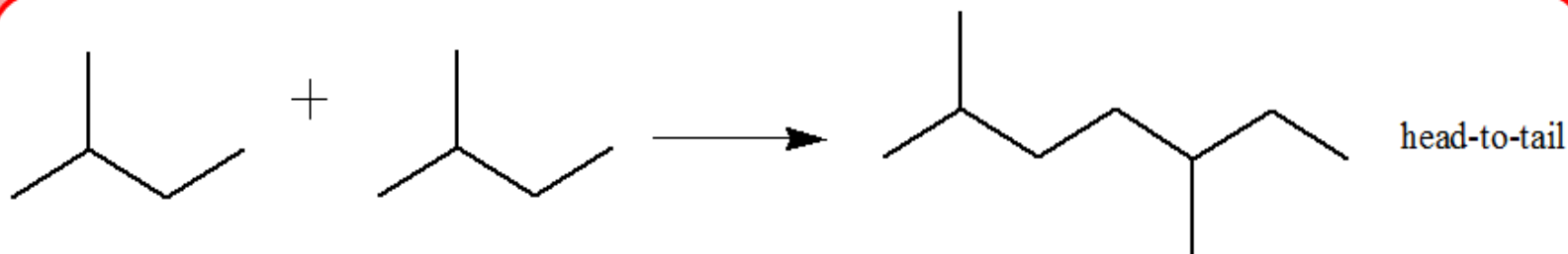
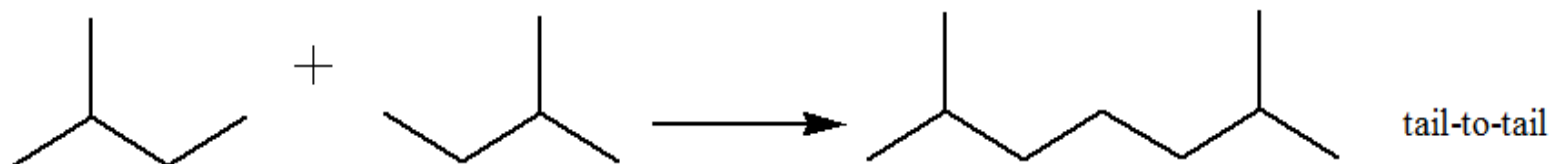
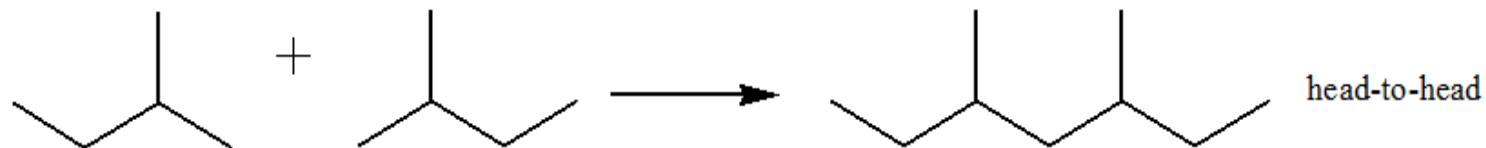
“terpenoids as chemicals containing a carbon skeleton formed by the joining together of isoprene units”

— Isoprene:

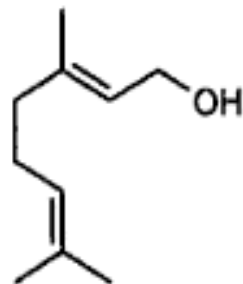


– Three possible connections are shown below:

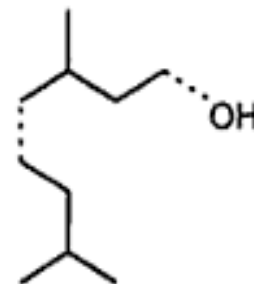
✓ Head-to-tail is the most common one



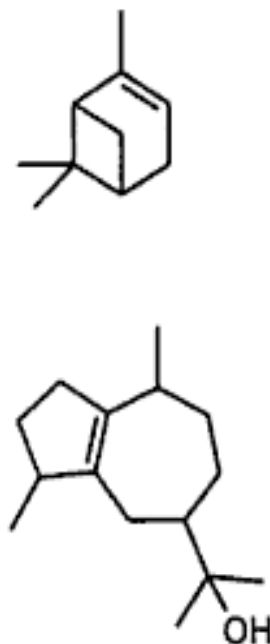
- Examples:



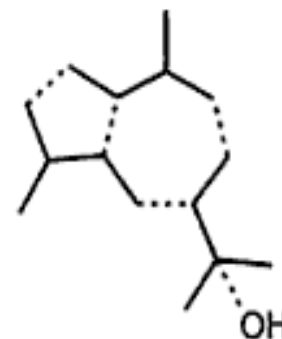
geraniol



α -pinene

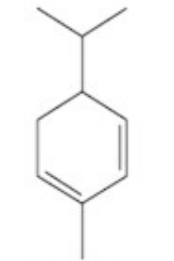


guaiol

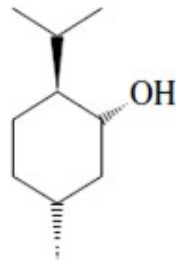


• Exercise 1:

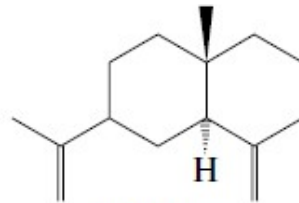
– Show the isoprene units of the following terpenoids



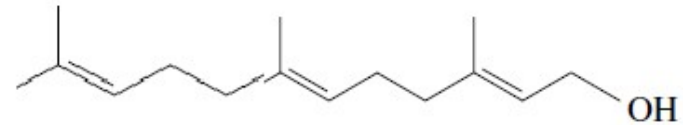
α -Phellandrene
(eucalyptus)



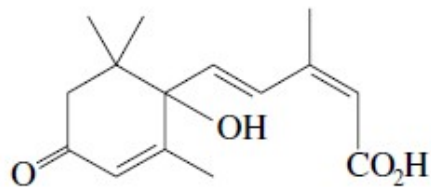
Menthol
(peppermint)



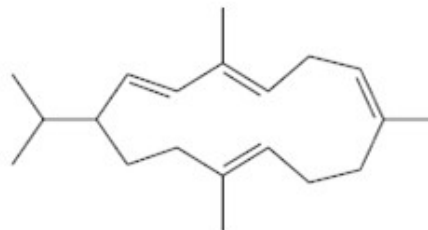
α -Selinene
(celery)



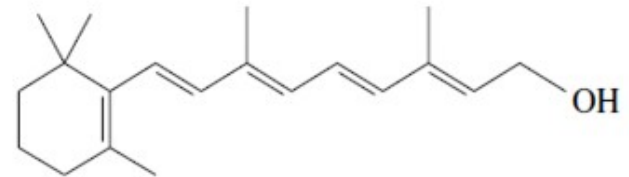
Farnesol
(ambrette)



Absciscic acid
(a plant hormone)

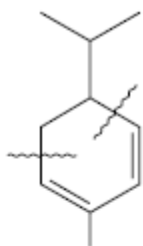


Cembrene
(pine)

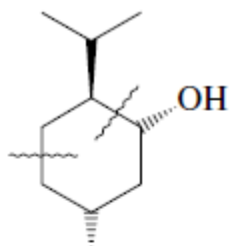


Vitamin A
(present in mammalian tissue and fish oil;
important substance in the chemistry of vision)

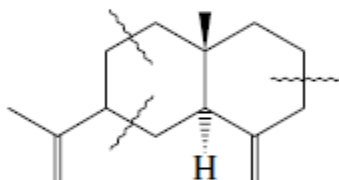
- **Answer 1:**



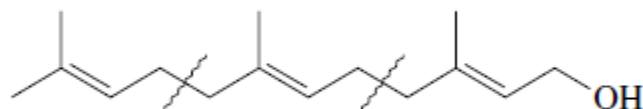
α -Phellandrene



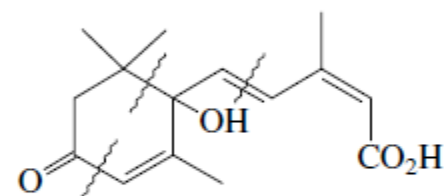
Menthol



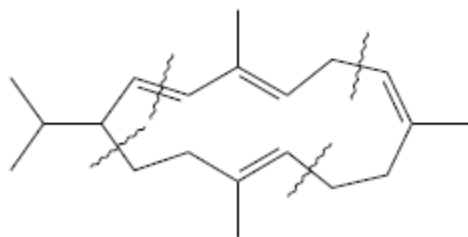
α -Selinene



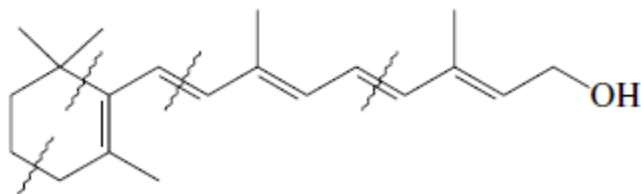
Farnesol



Absciscic acid

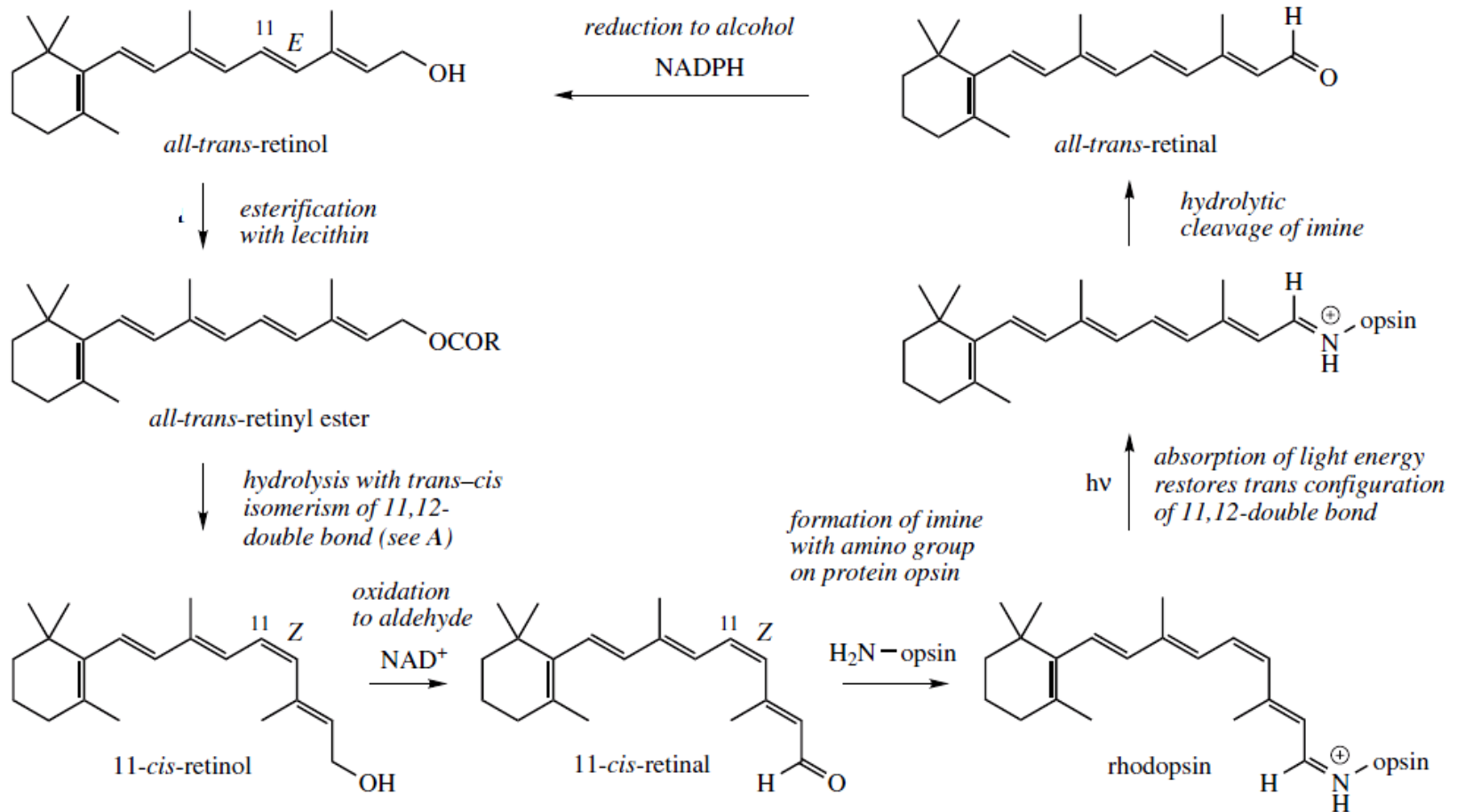


Cembrene



Vitamin A

- Chemistry of vision: role of vitamin A



Terpenoid nomenclature

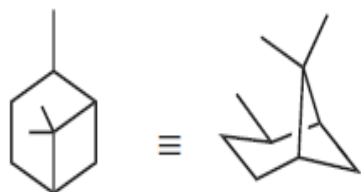
- **Terpene nomenclature**
 - **Systematic + trivial + semi-systemic name**
 - **Basic skeletons**

MONOCYCLIC

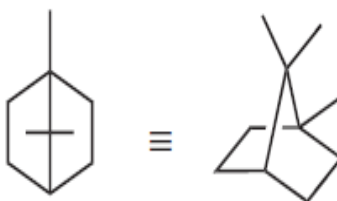


menthane
type

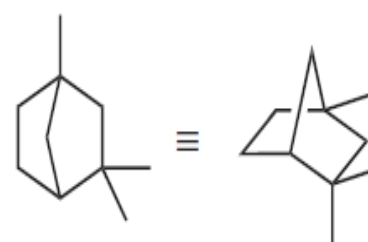
BICYCLIC



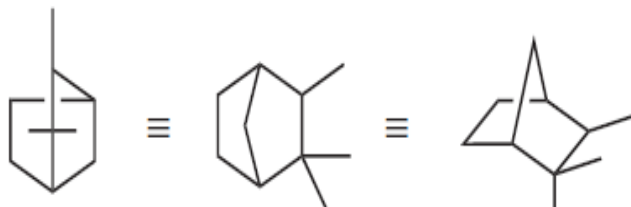
pinane type



camphane/bornane
type



fenchane type



isocamphane type



carane type

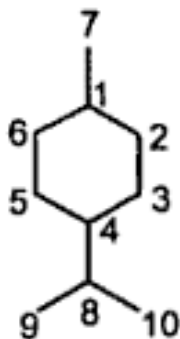


thujane type

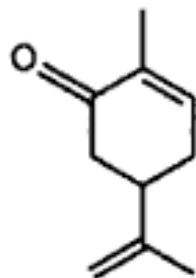
- **Example:**

- Carvone

- Carvone
 - *p*-mentha-1,8-dien-6-one
 - 1-methyl-4-(1-methylethenyl)-cyclohex-1-ene-6-one.



p-menthane

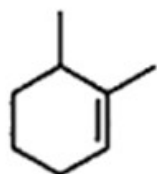


carvone

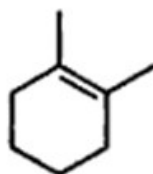


- **Greek letters**

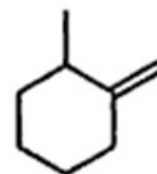
- To distinguish between isomeric terpenoids



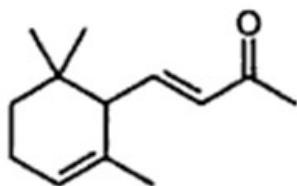
α -isomer



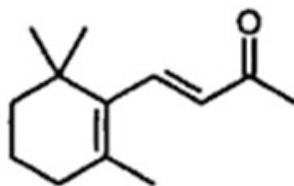
β -isomer



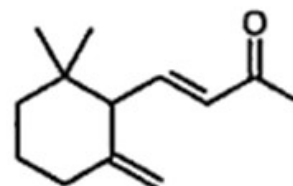
γ -isomer



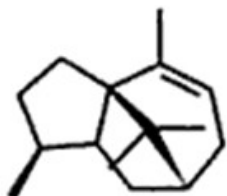
α -ionone



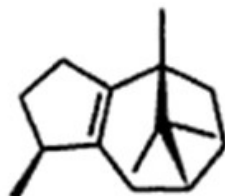
β -ionone



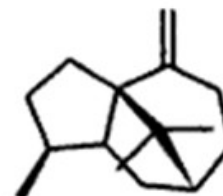
γ -ionone



α -patchoulene



β -patchoulene



γ -patchoulene

Significance of terpenoids in pharmacy

- **In animals terpenoids are important as**
 - **component of cell membrane**
 - **hormones**
 - **pheromones**

- **In plants**
 - **flavors and scents (pollinator attractants)**
 - **growth regulators**
 - **defense secretions (e.g. phytoalexins)**

- **Economical & commercial importance**
 - In agriculture: important biodegradable insecticides and herbicides e.g. gibberillins
 - In perfumery & flavor production

- **In medicine**
 - hypercholesterolemia e.g. compactin
 - for treatment of cancer e.g. paclitaxel (taxol)
 - for treatment of helmethiasis e.g. α -santonin
 - antiprotozoal
 - » against leishmaniasis e.g. amarogentin
 - » against malaria e.g. artemisinin
 - » against trypanosomiasis
 - antimicrobial: due to detrimental effect on the cell membrane
 - contraceptive:
 - » as raw material for semisynthesis
 - » directly e.g. gossypol
 - biodegradable insecticides e.g. pyrethrins
 - artificial sweeteners e.g. stevioside

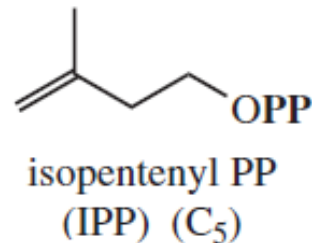
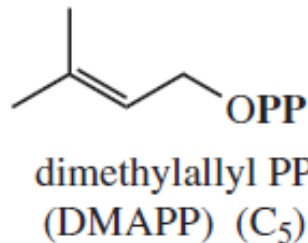
- **Terpenes**

- Attraction of both pollinators and predators
- Protection against photooxidative stress
- Mediating thermotolerance
- Direct defense against microbes and insects

Biosynthesis of terpenes:

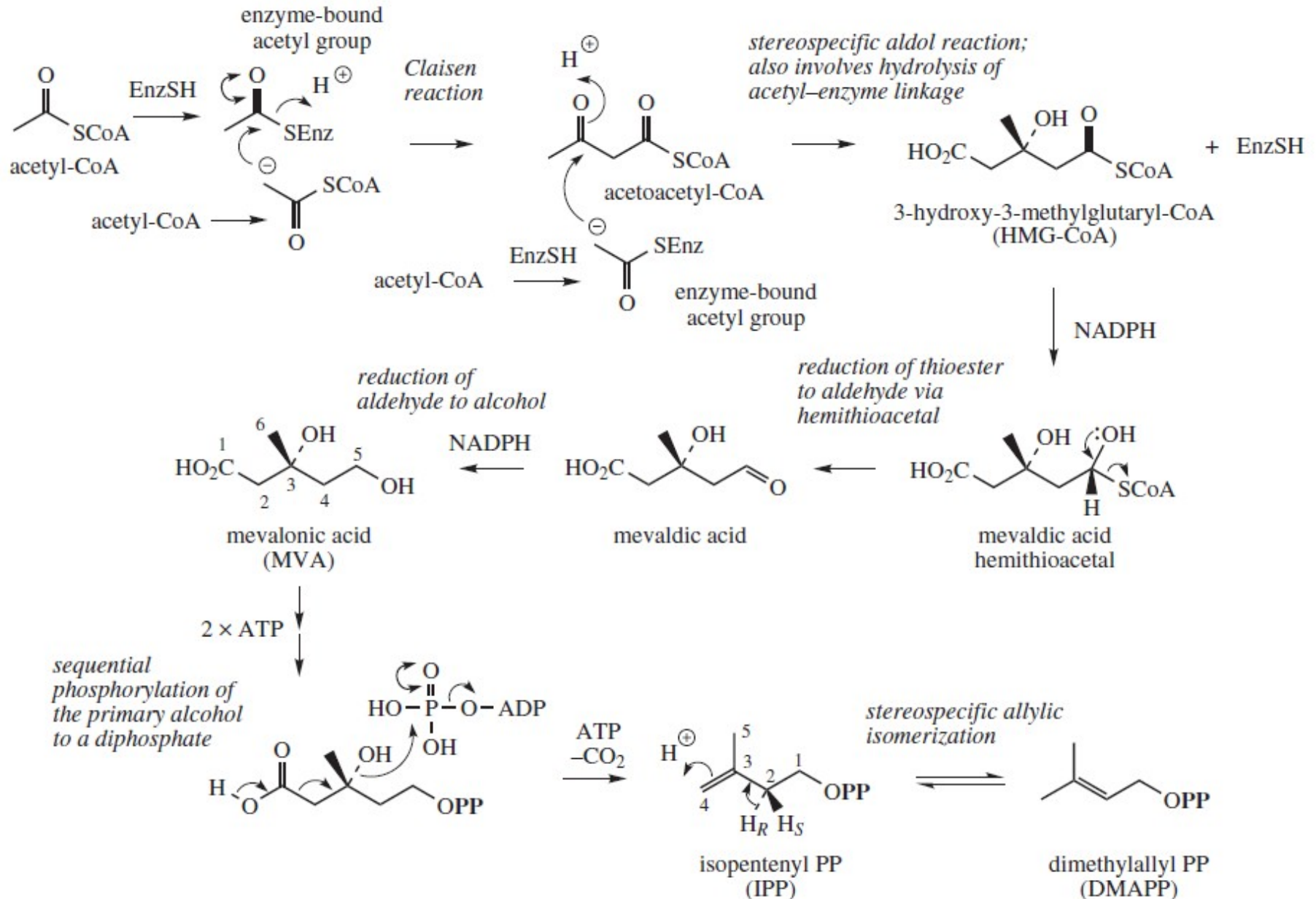
An overview

- From Isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP)

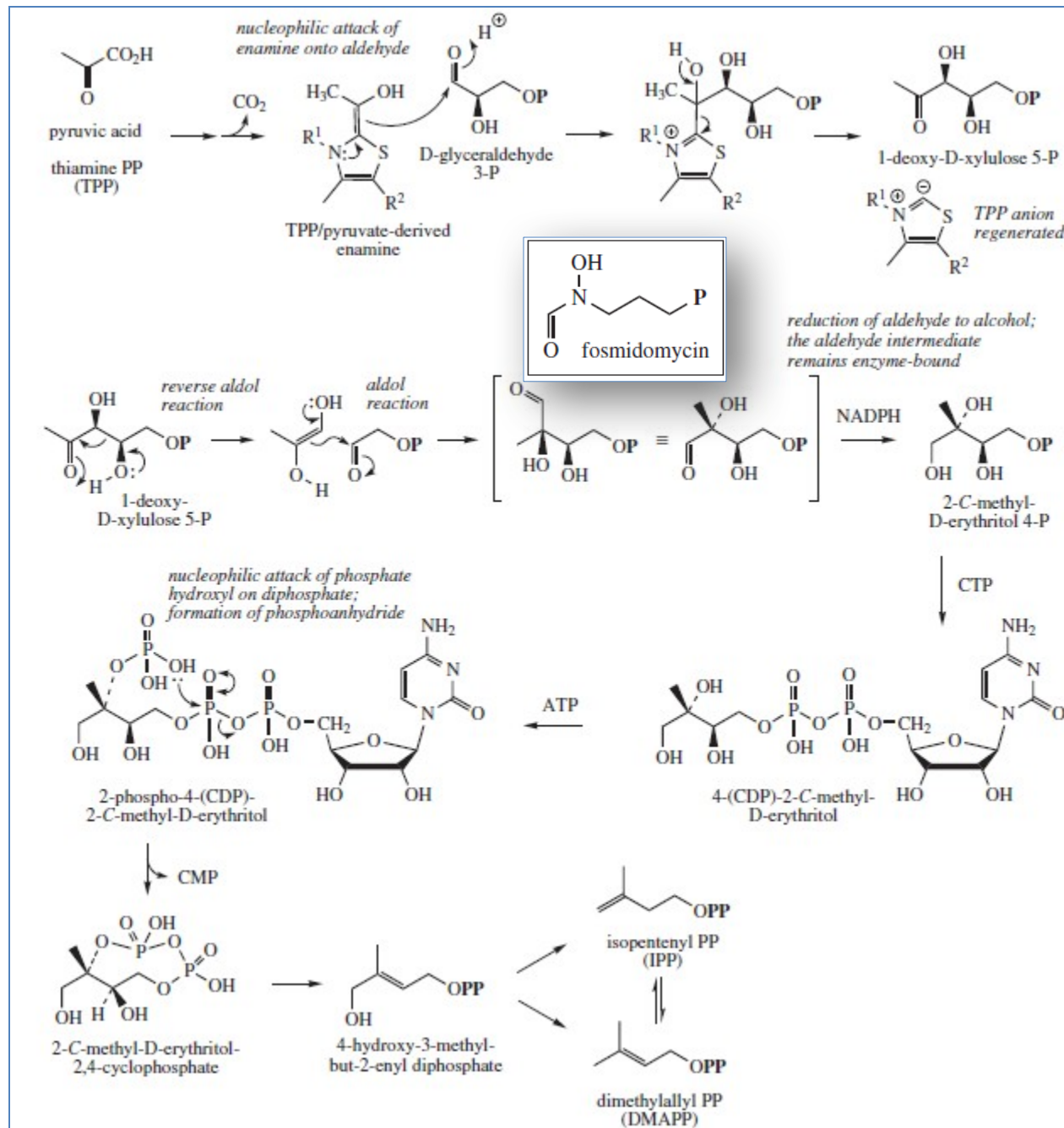


- Two pathways:
 - The *mevalonic-acid (MVA) pathway*
 - The *2-C-methylerythritol 4-phosphate (MEP) pathway*

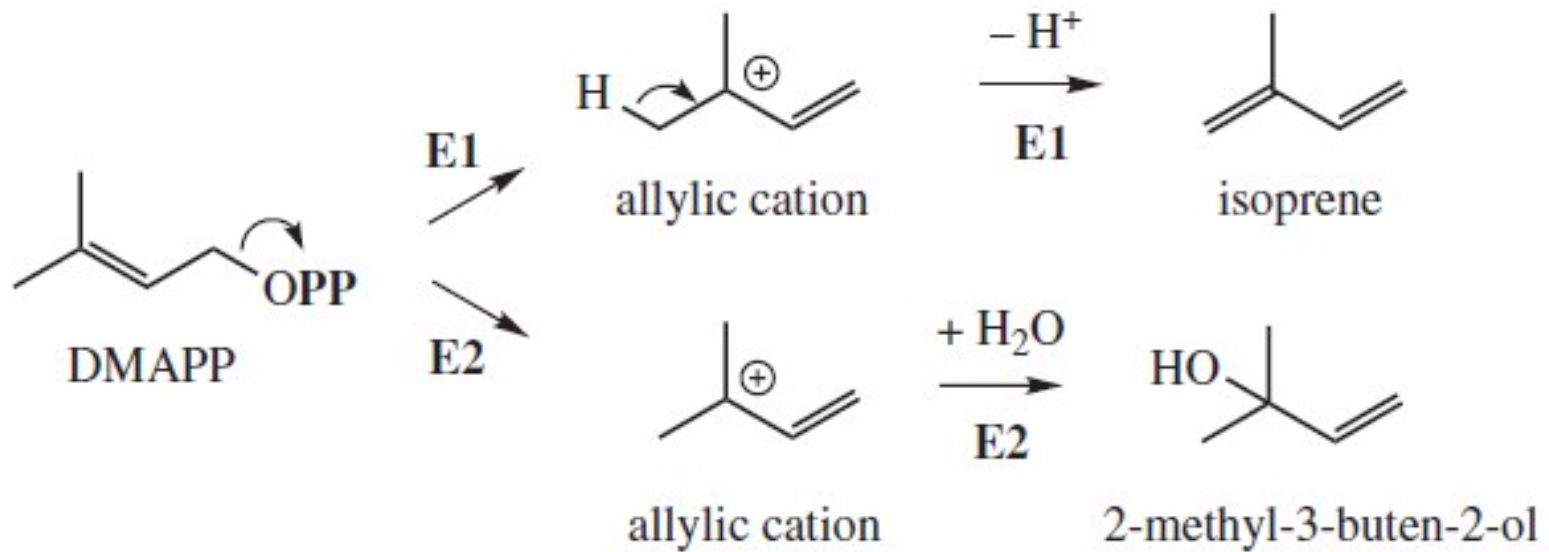
The *mevalonic-acid (MVA) pathway*



The *MEP* pathway



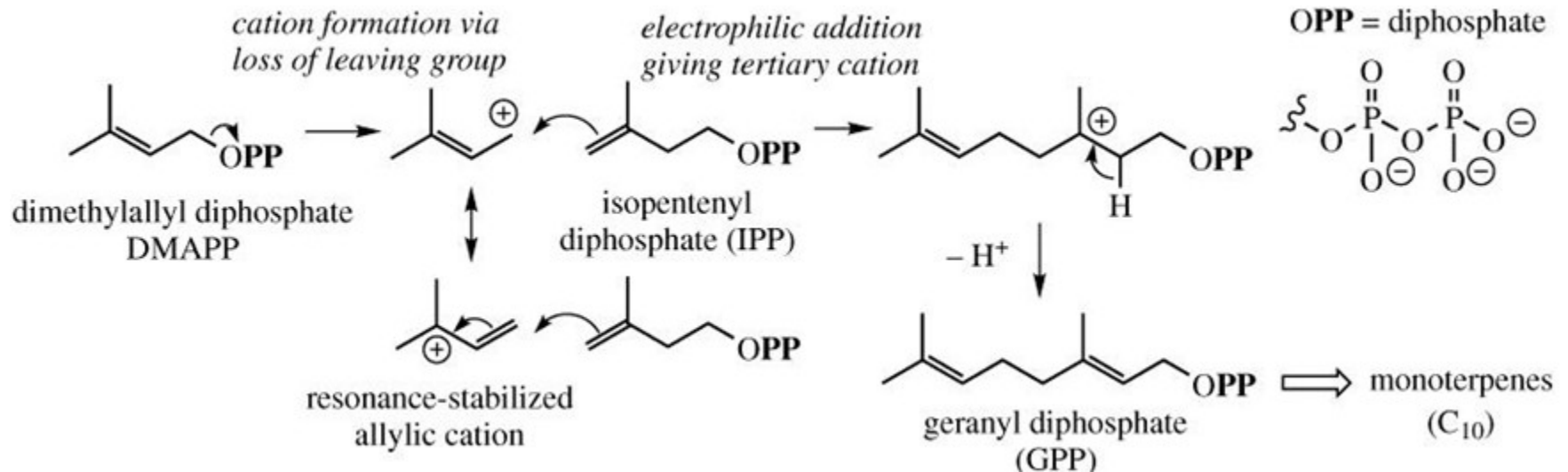
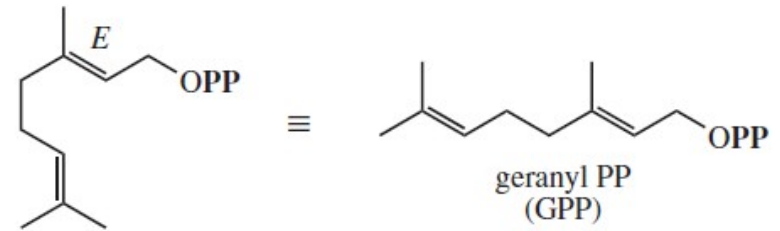
- **Hemiterpenes (C5)**



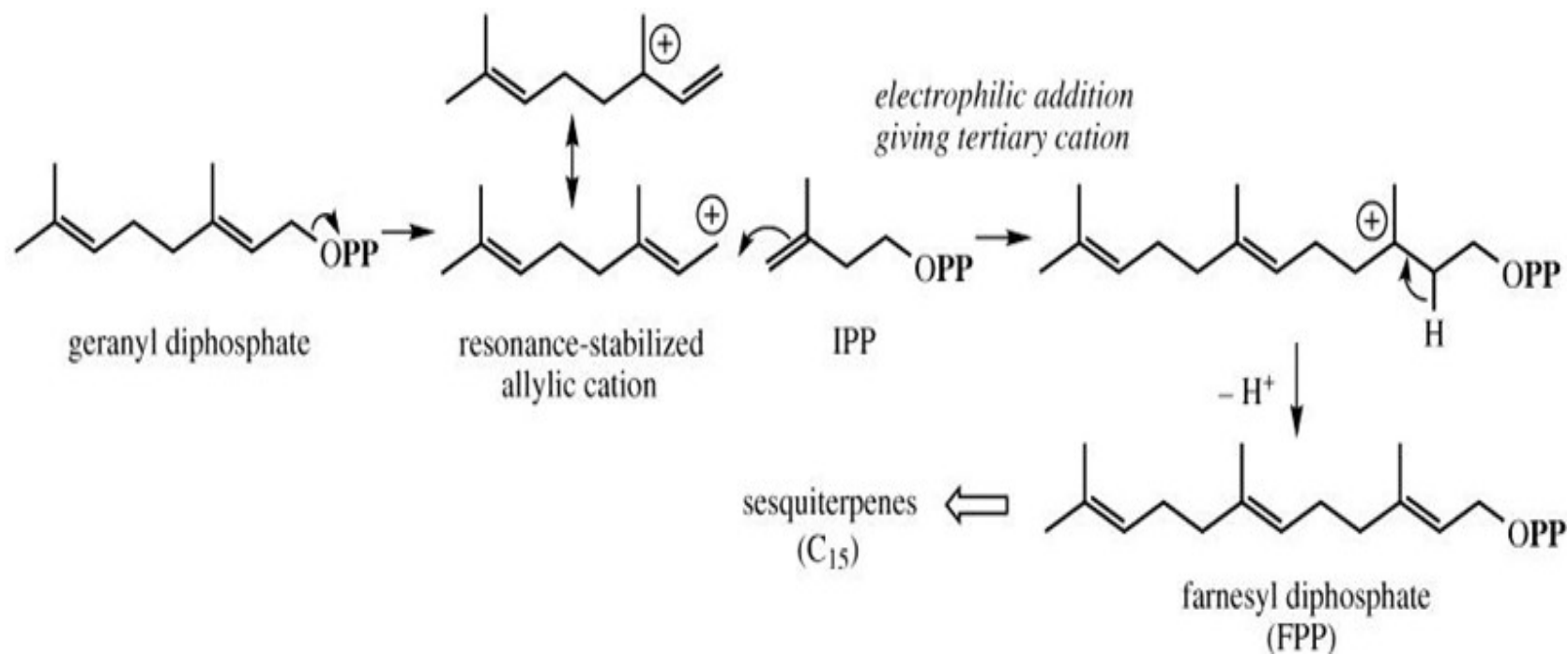
E1: isoprene synthase

E2: methylbutenol synthase

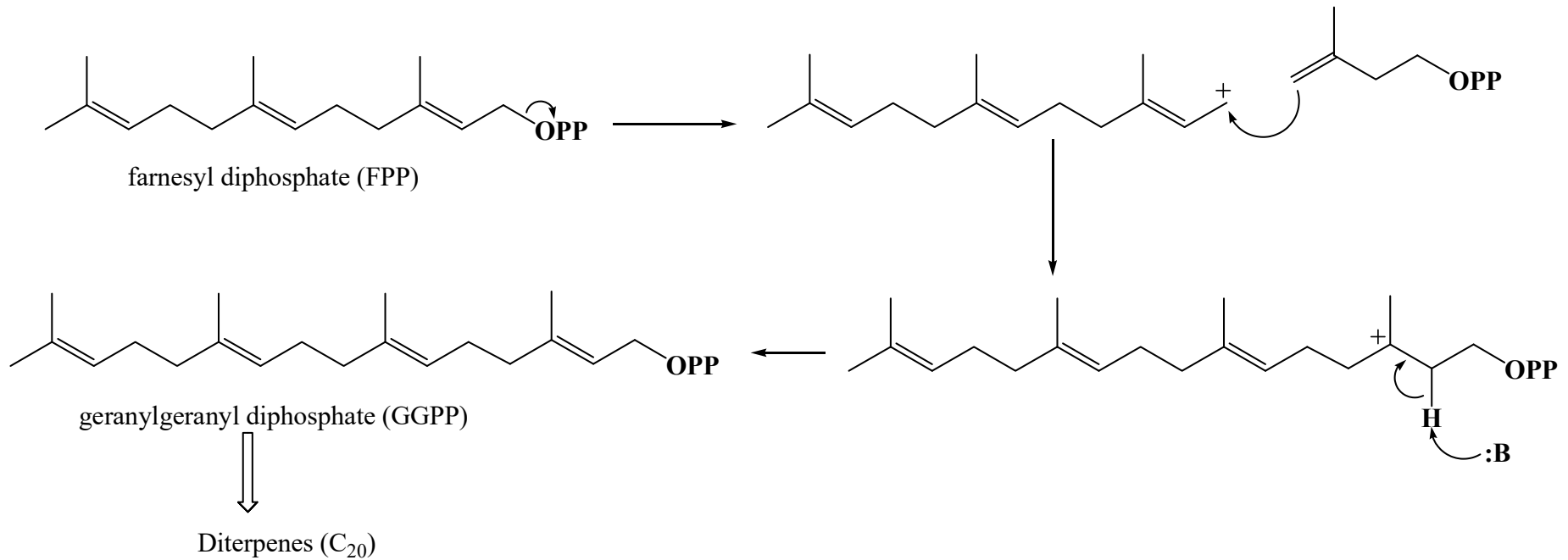
- **Prenyl diphosphates**
 - Geranyl diphosphate (GPP)



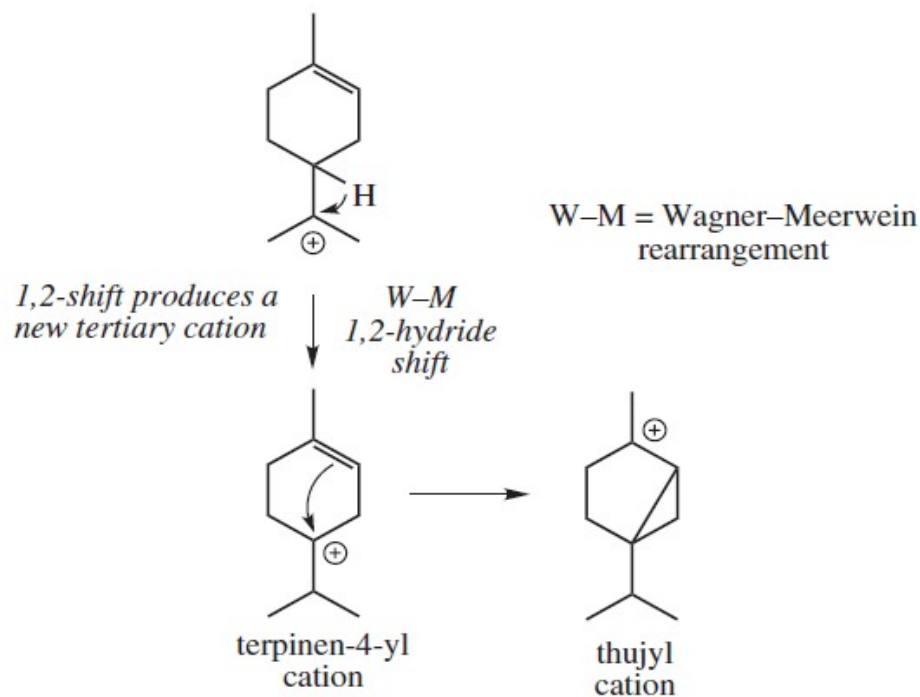
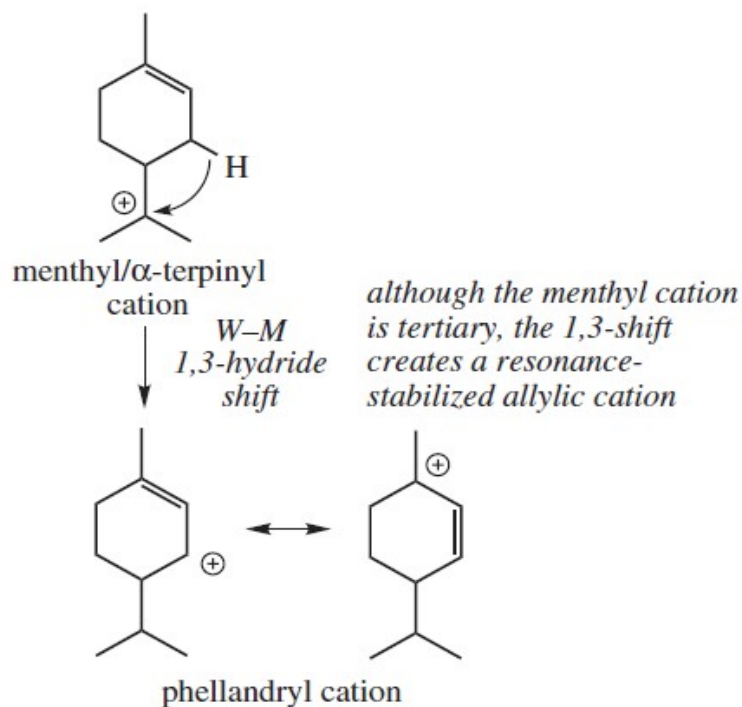
- **Farnesyl diphosphate**



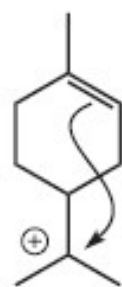
- **Geranylgeranyl diphosphate (GGPP)**



- **Diversity**
 - Terpene synthases/cyclases (TPSs)
 - Carbocations: three main reactions
 - **Hydride shifts**



- Alkyl shifts: Wagner–Meerwein rearrangements



*electrophilic addition
creates 4-membered
ring and tertiary
carbocation*

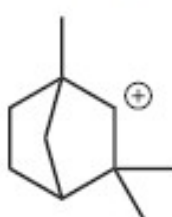


*1,2-shift creates secondary
cation, but reduces ring strain*

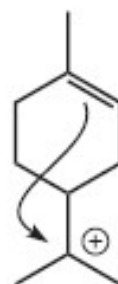


pinyl
cation

*W-M
1,2-alkyl
shift*



fenchyl
cation



*electrophilic addition
creates 5-membered
ring and secondary
carbocation*

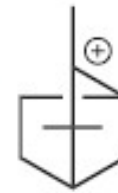


*1,2-shift creates
tertiary cation*



bornyl
cation

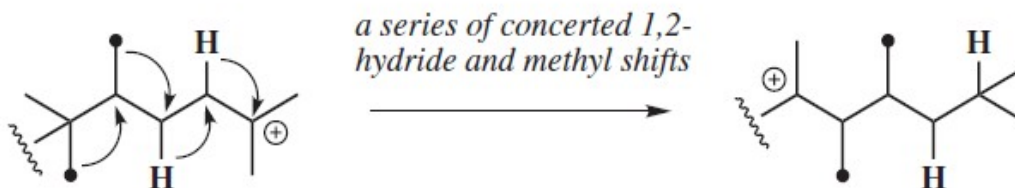
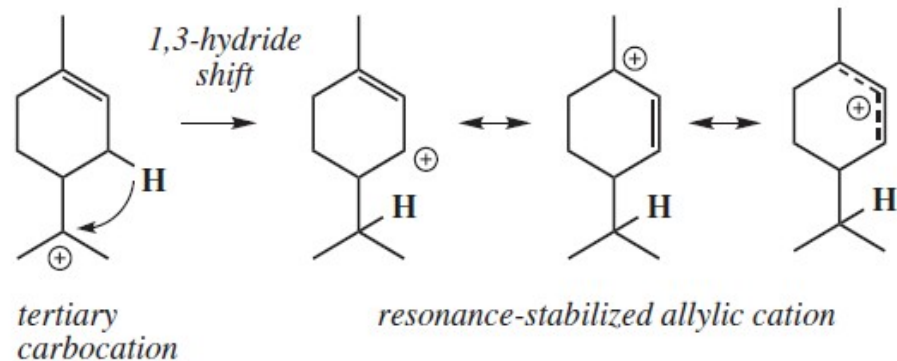
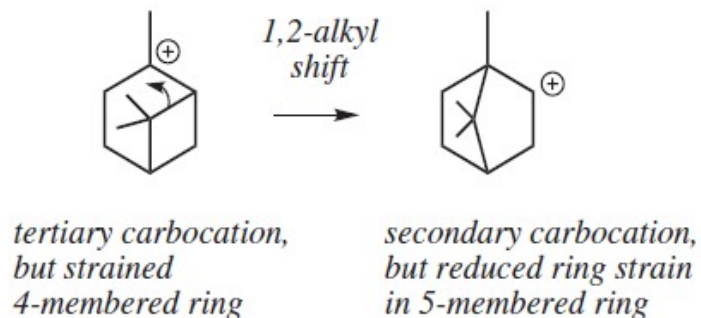
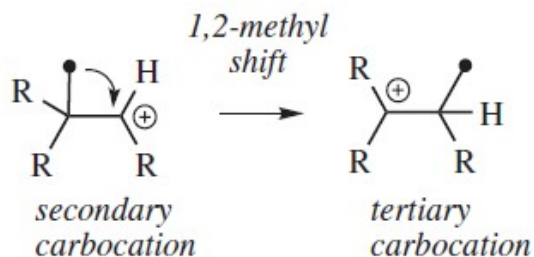
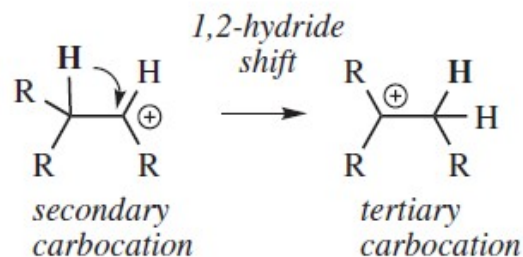
*W-M
1,2-alkyl
shift*



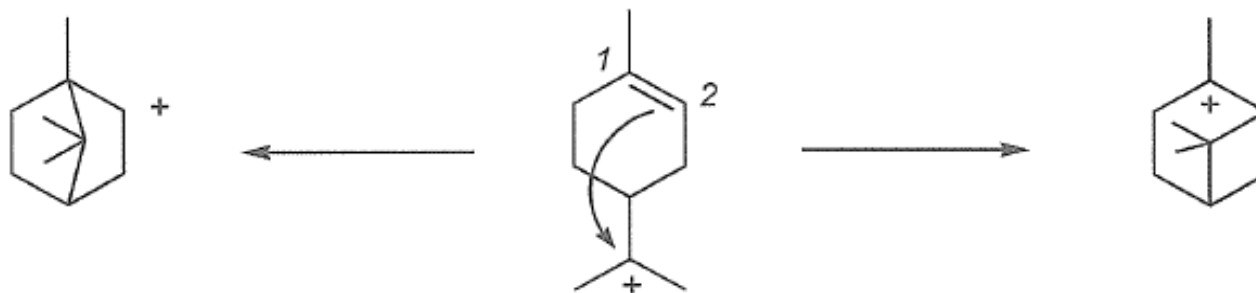
isocamphyl
cation

• Wagner–Meerwein rearrangements

Wagner–Meerwein rearrangements

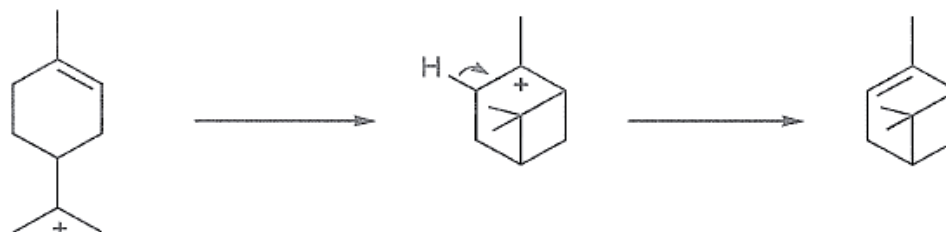


– Cyclizations

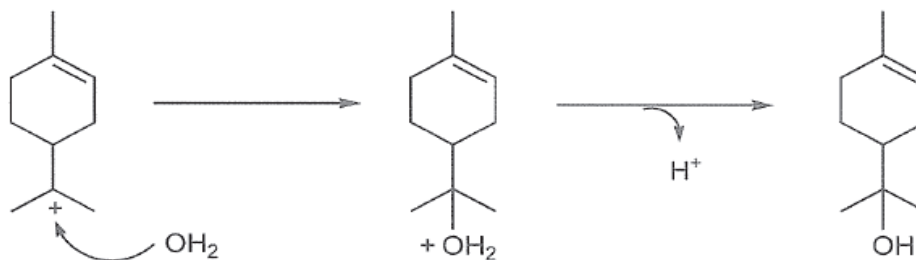


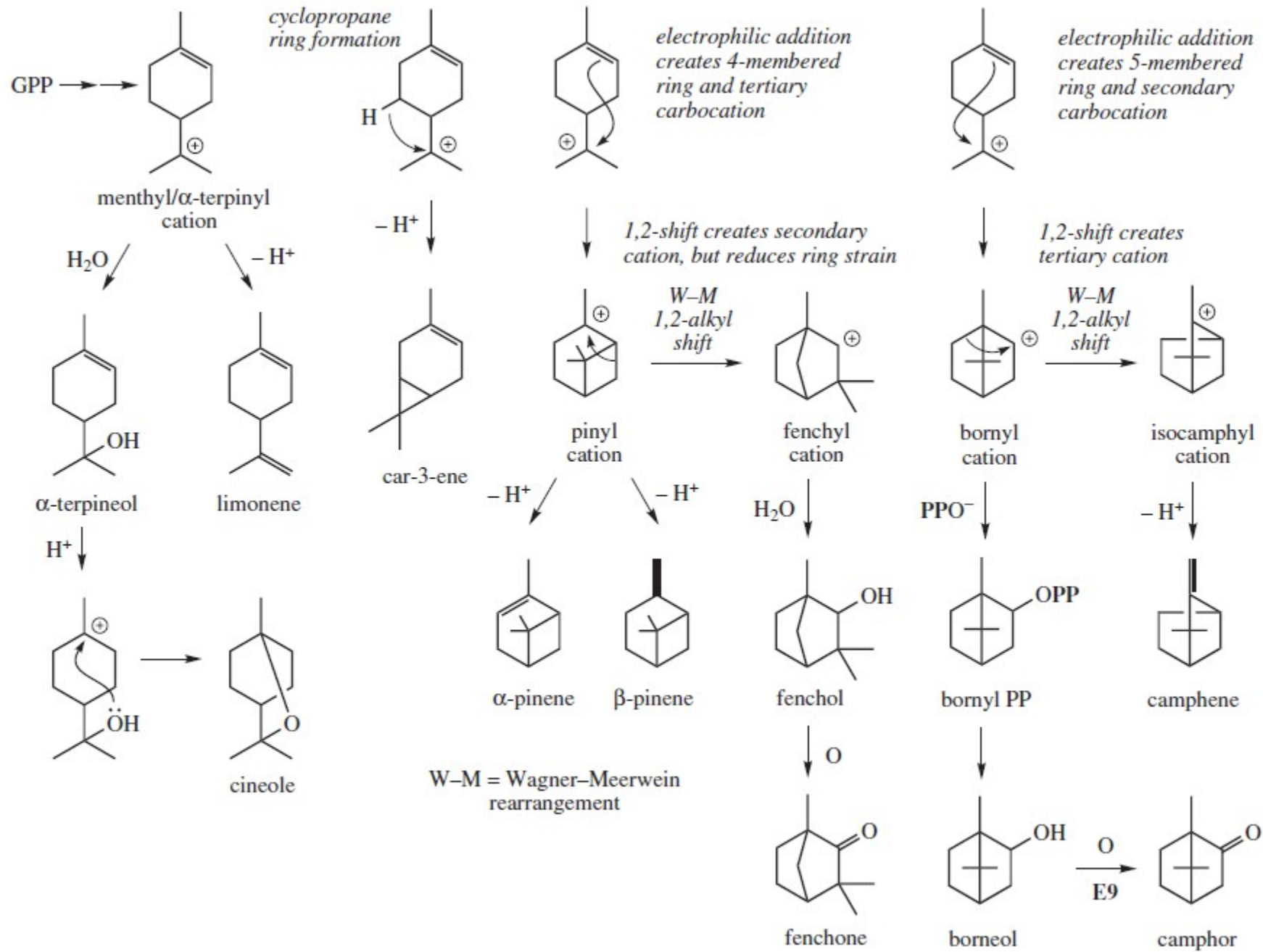
• Termination of carbocations:

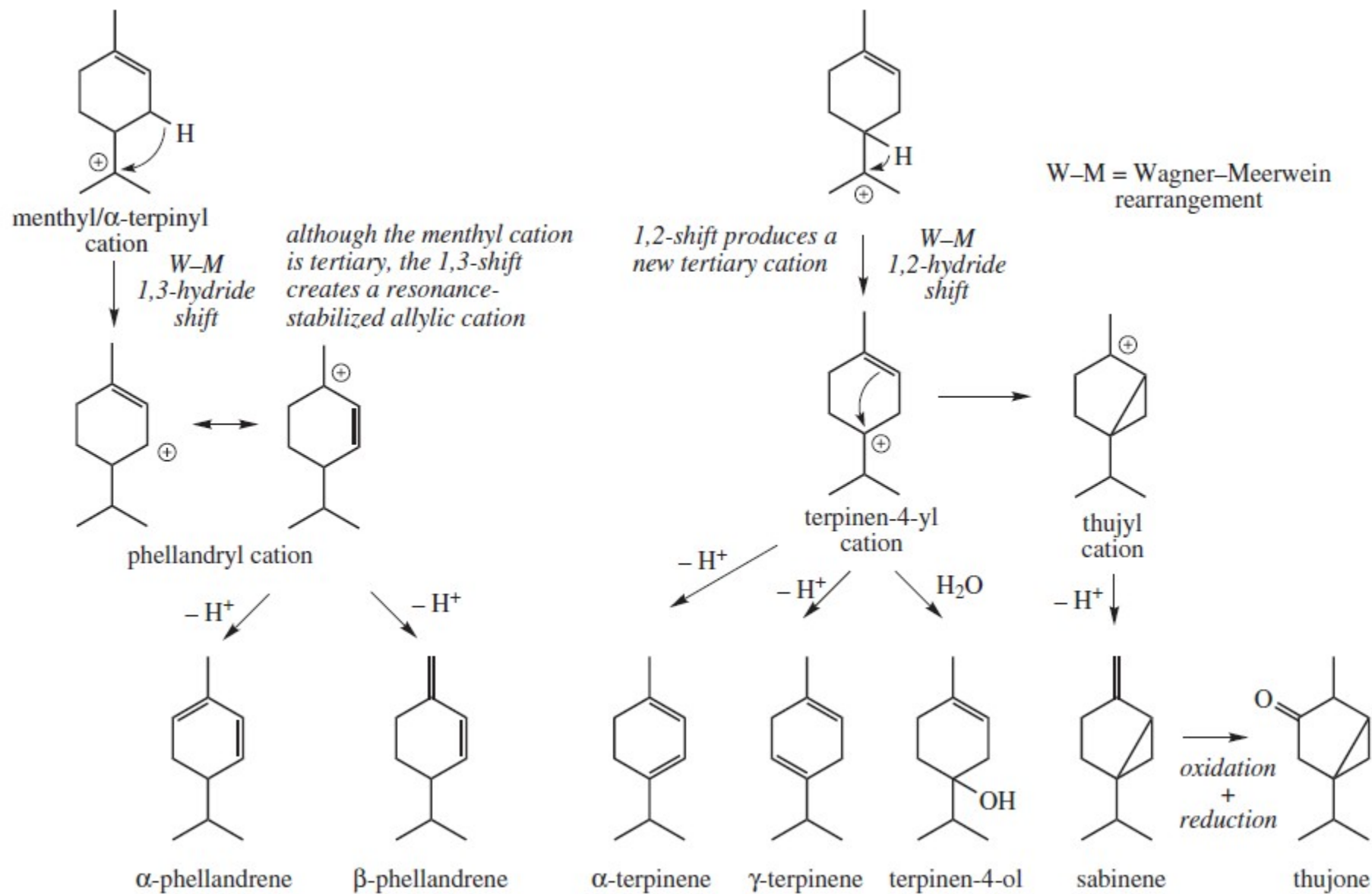
– Loss of a proton



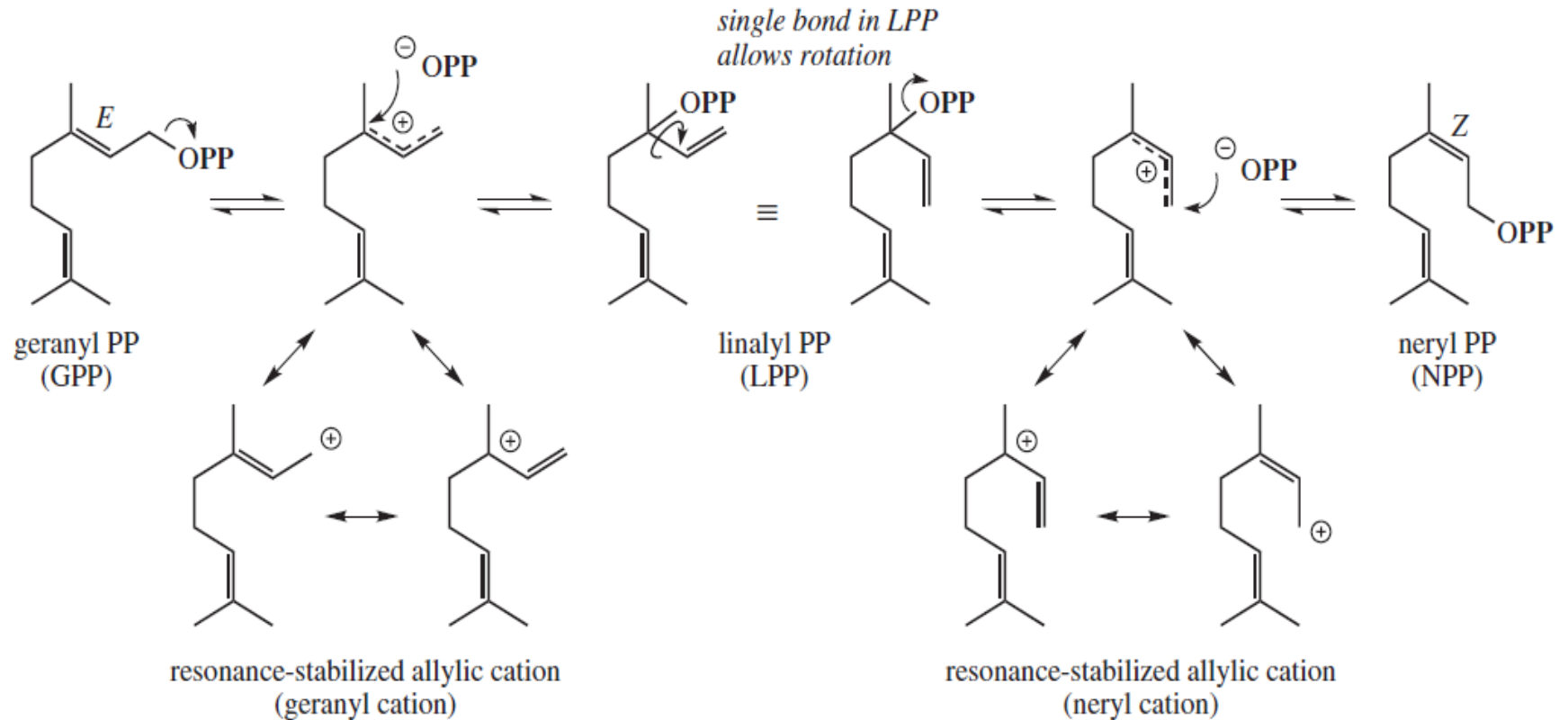
– Addition of water followed by loss of a proton



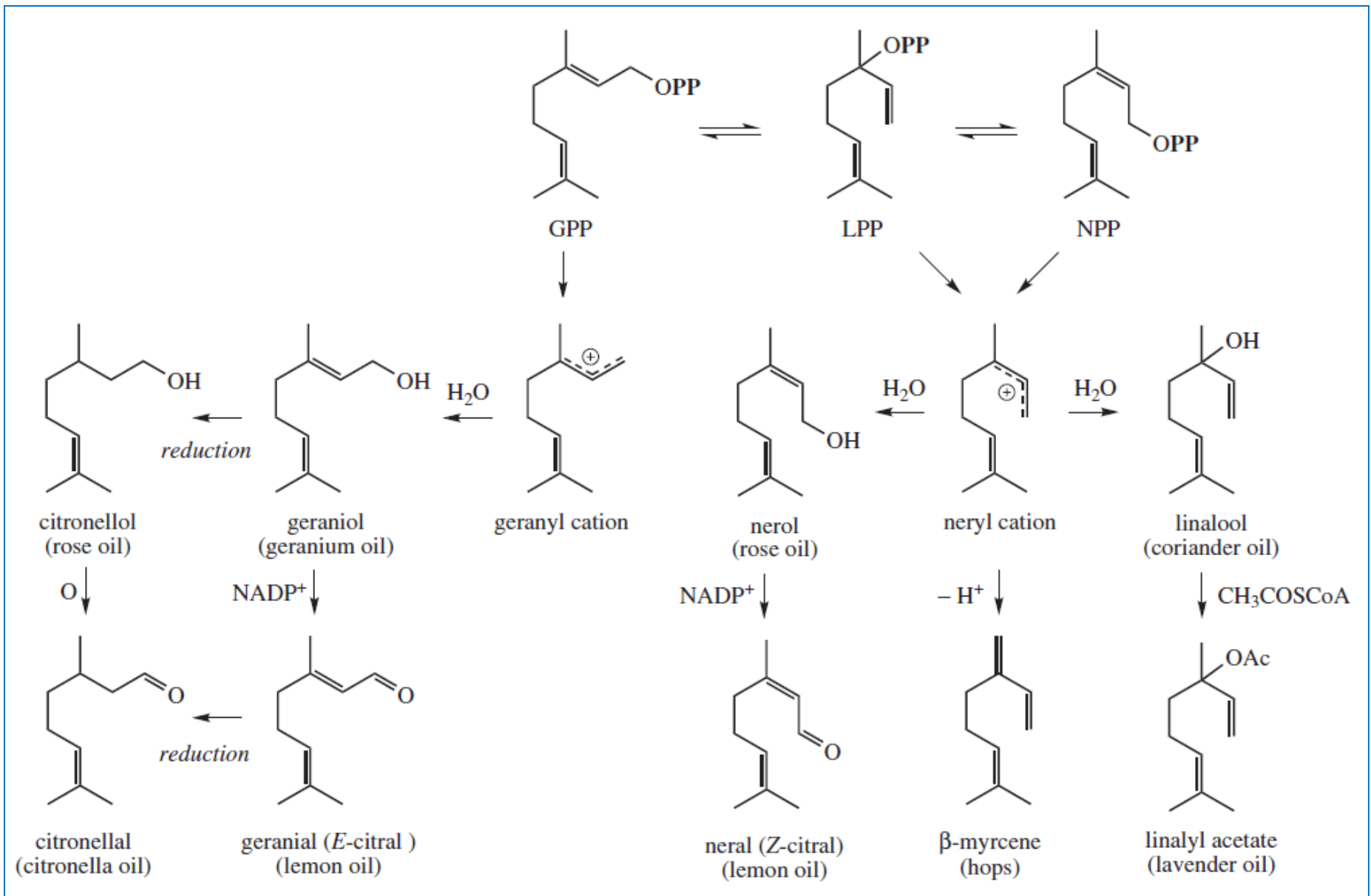




- **Linalyl PP (LPP) and neryl PP (NPP)**

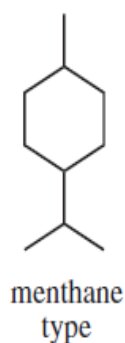


- **GPP, LPP and NPP $\Rightarrow$ linear monoterpenes**

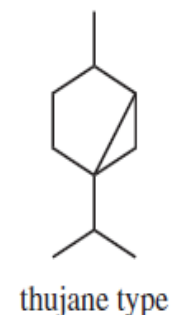
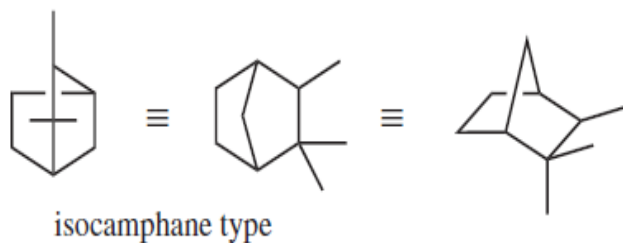
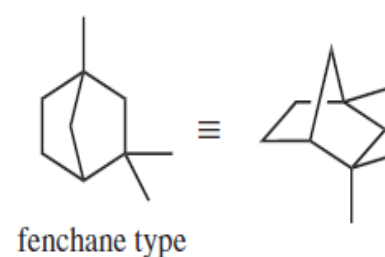
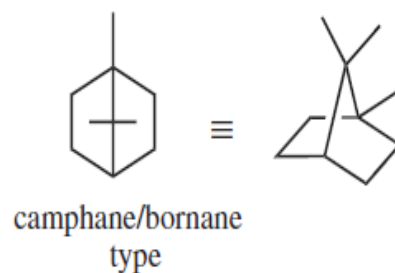
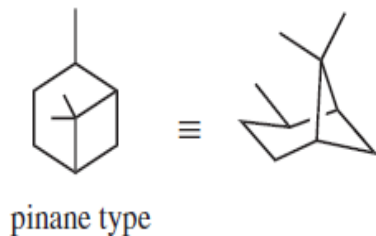


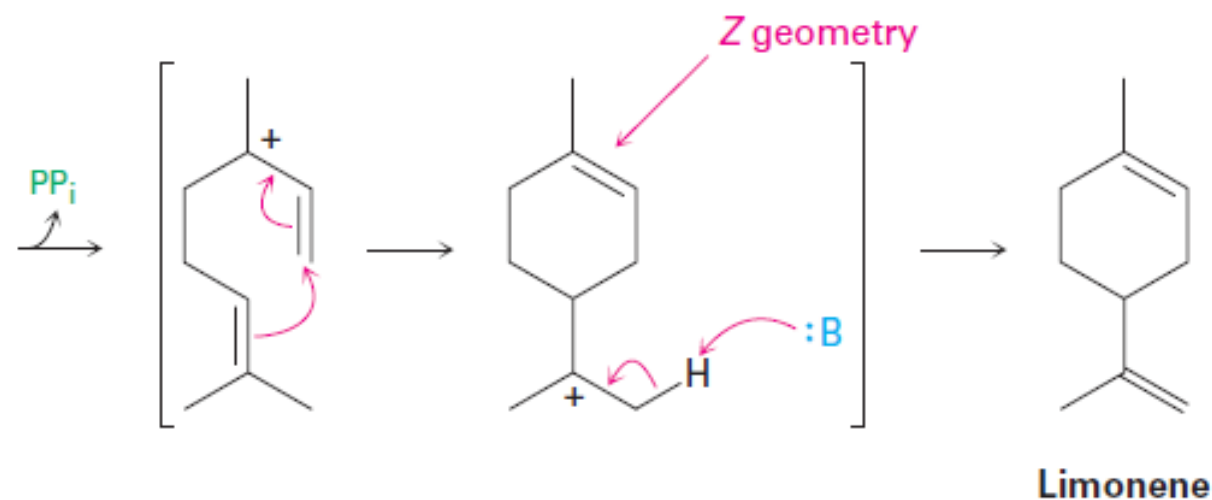
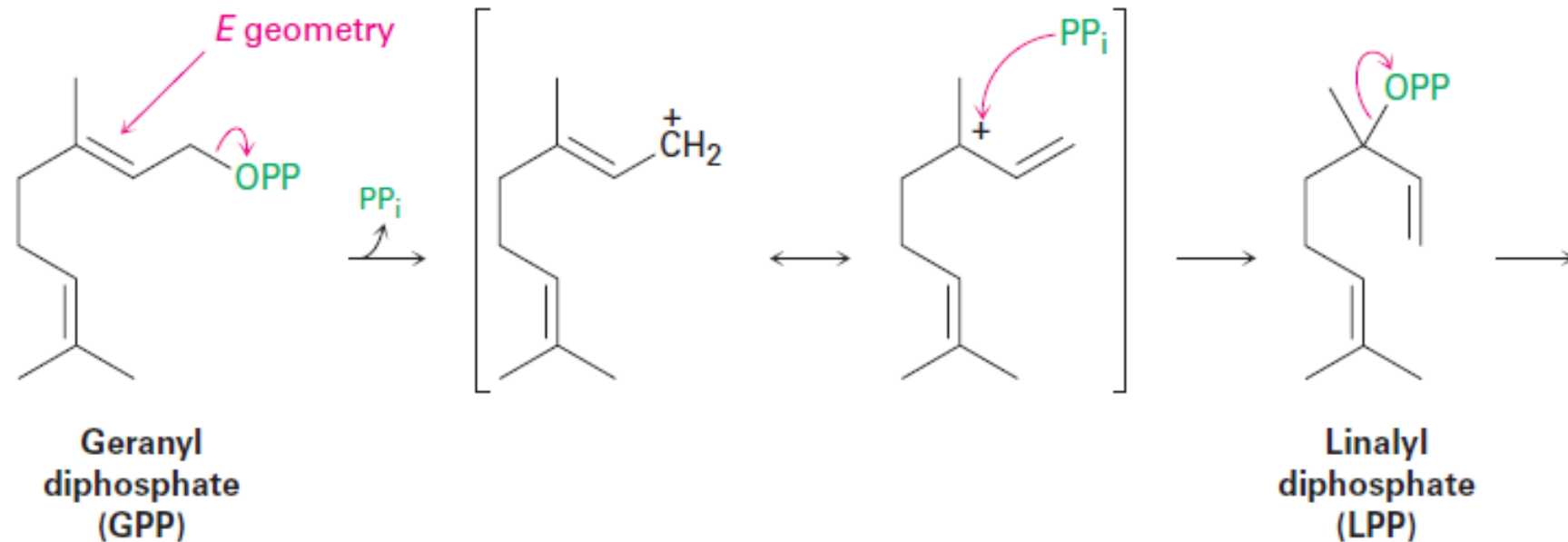
- **Cyclic monoterpenes**
 - Intramolecular electrophilic addition

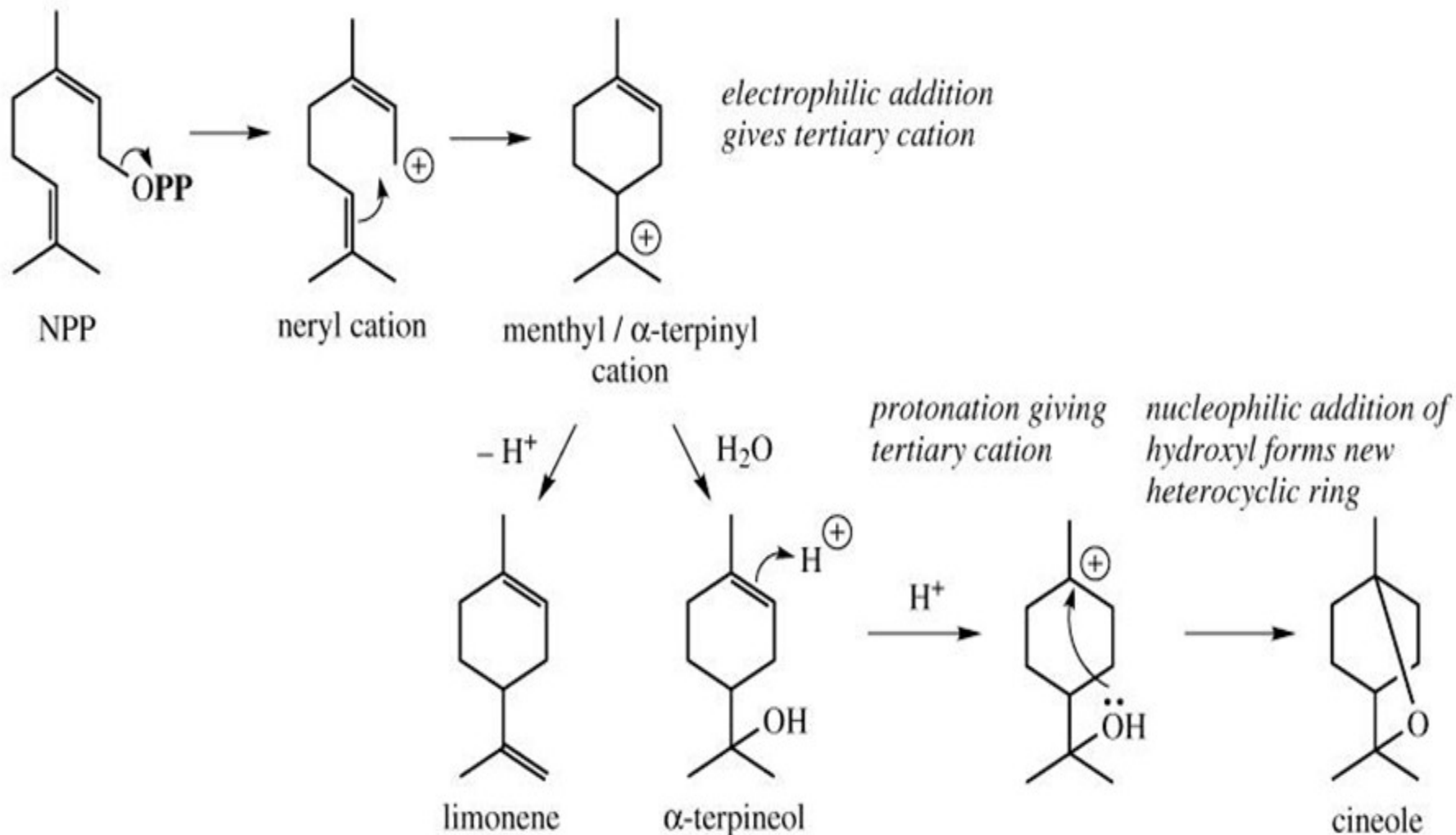
MONOCYCLIC



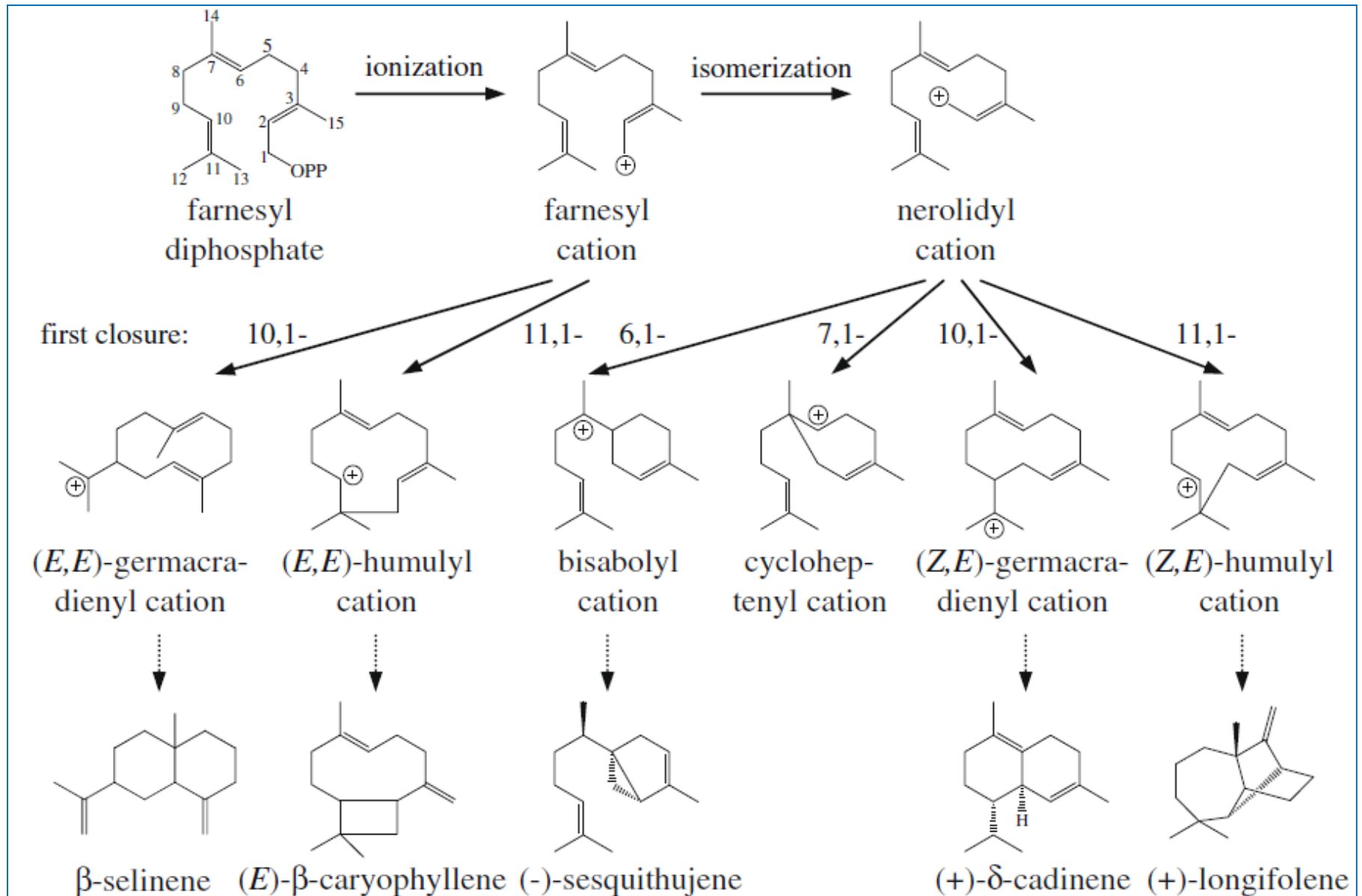
BICYCLIC

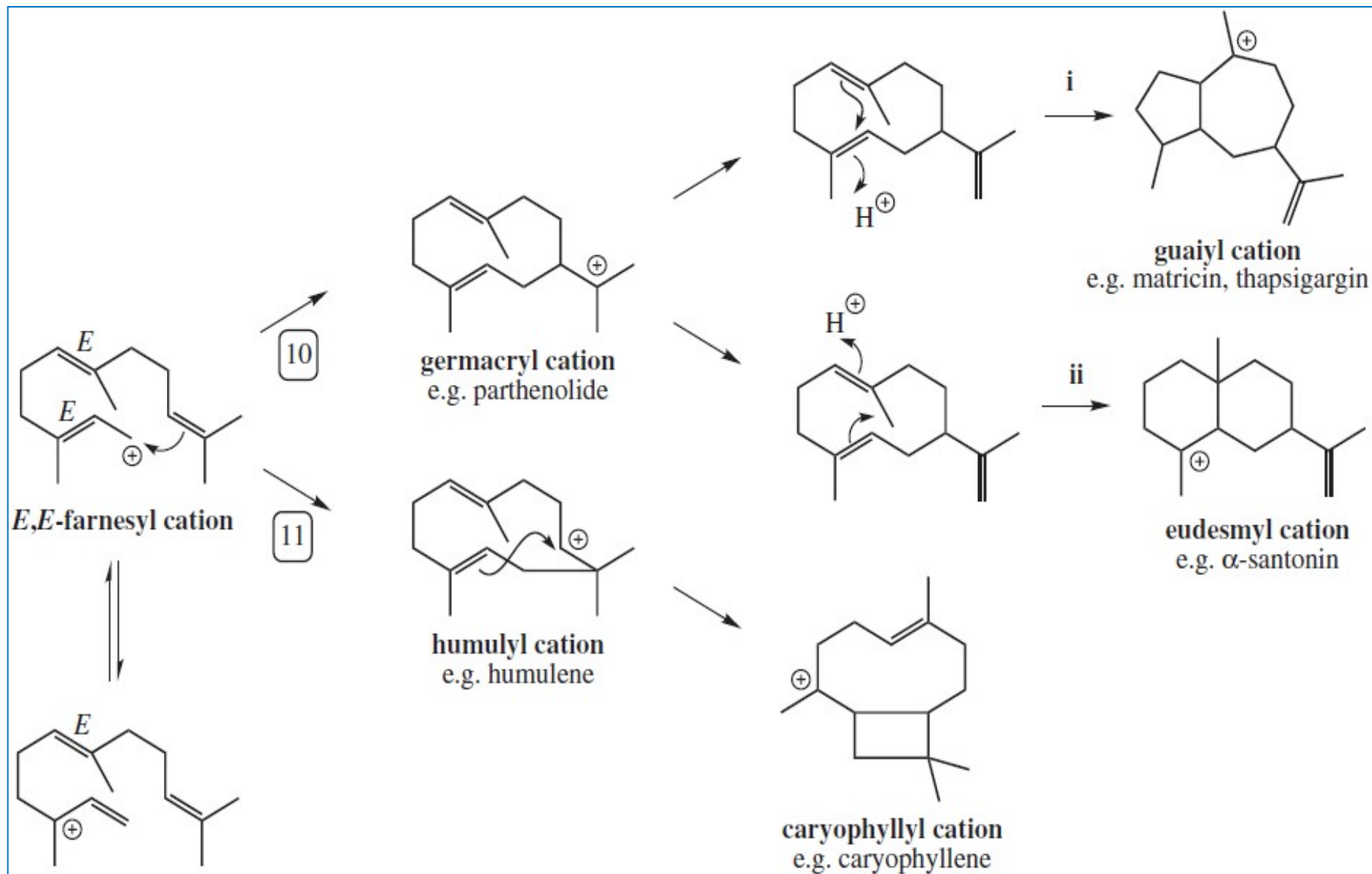


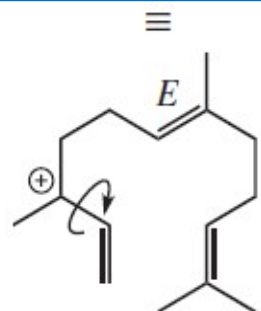




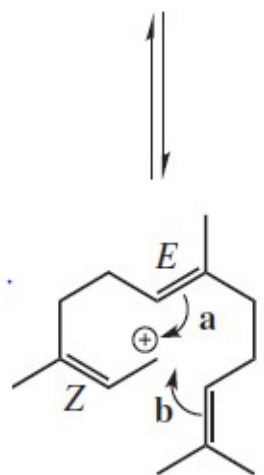
- **Cyclic sesquiterpenes from FPP**



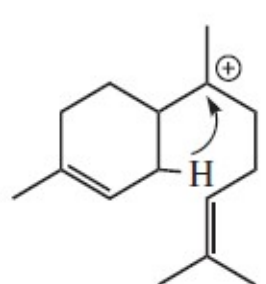
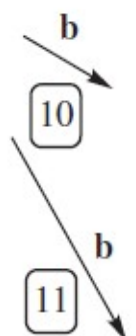
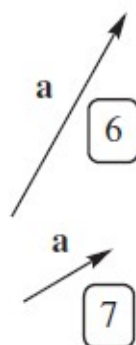




nerolidyl cation

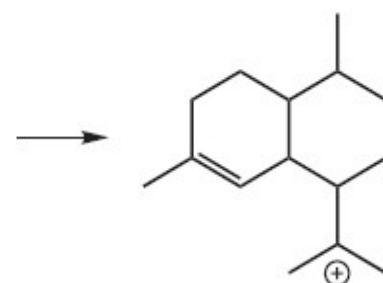
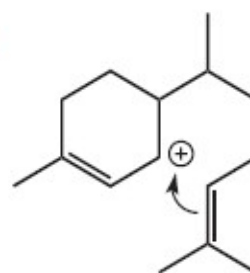


E,Z-farnesyl cation

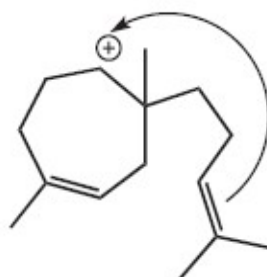


bisabolyl cation
e.g. bisabolene, α -bisabolol

W-M
1,3-hydride
shift

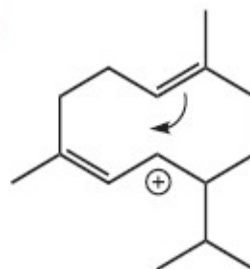


amorphyl cation
e.g. artemisinic acid

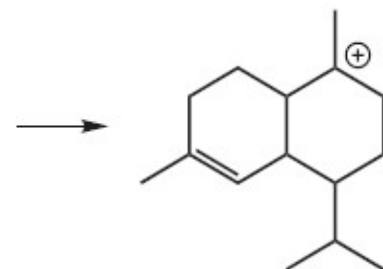


carotyl cation

W-M
1,3-hydride
shift

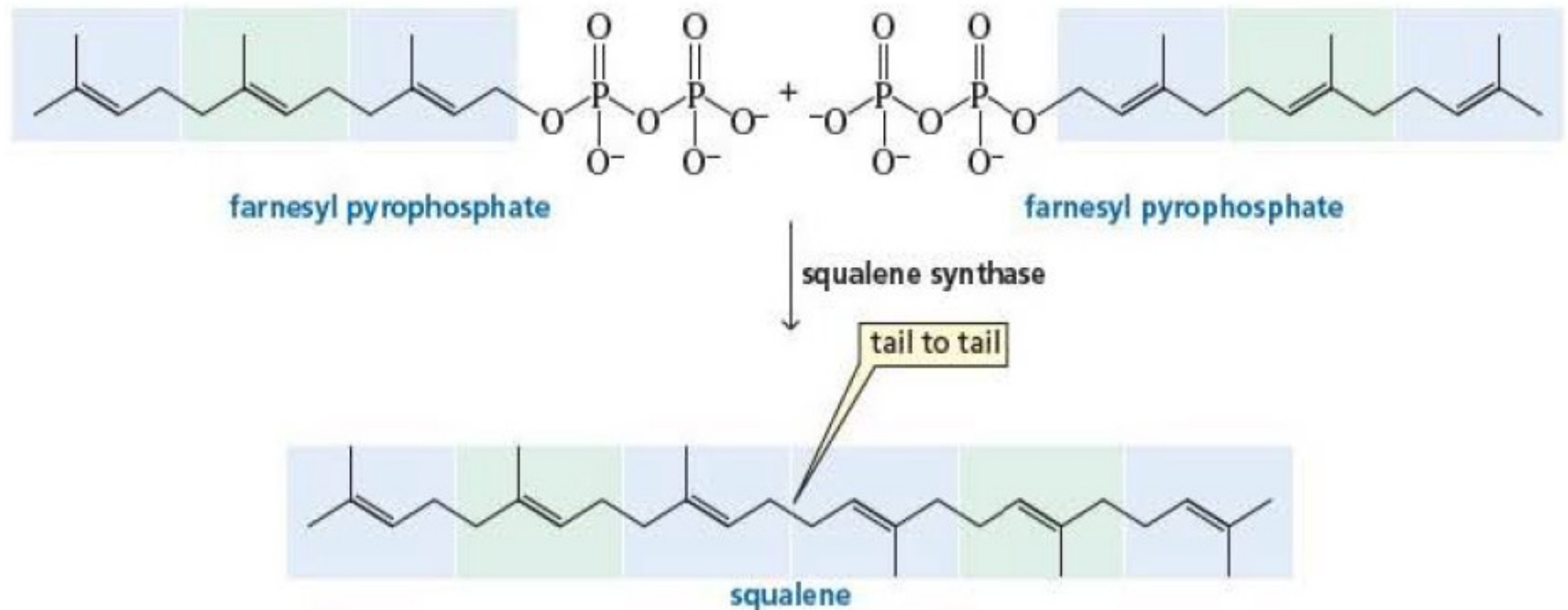


cis-germacryl cation



cadinyl cation
e.g. α -cadinene

- **Triterpenoids (C30) and tetraterpenoids (C40)**
 - Tail-to-tail dimerization of C15 and C20 units



- **Summary**

